important builders of modern India, as revealed by their correspondence. Mahatma Gandhi's motto was "industrialize and perish," while M. Visvesvaraya's was "industrialize or perish," and in the 1930s Mahatma Gandhi had written to M. Visvesvaraya that "in India, at any rate for generations to come, we shall not be able to make much use of mechanical power for solving the problem of the ever growing poverty of the masses."[77]

M. Visvesvaraya replied:

> You say we hold perhaps diametrically opposite views. You are for developing village industries and I favor both heavy industries and village industries. To the extent that you propose to advance village industries, I am at one with you. I can never persuade myself to take up a hostile attitude towards any constructive work, from any quarter … I am in favor of heavy industries because heavy industries will save the money that is going out of the country in large sums every year; heavy industries are required to provide the local manufactures of machinery and equipment required by our railways and for defence forces and heavy industries are required also for supplying machinery and tools for the village industries themselves. I recommend more extended use of mechanical power because it produces results for the country much more rapidly than human power. The object is to get food and commodities required by our people for a decent standard of living as speedily as possible.[78]

Fortunately, there is no longer any controversy between Bombay and Calcutta schools of thought. India, as a whole, has emerged as a nation in M. Visvesvaraya's vision. Indian industry is presently ideology free. Pragmatism and native entrepreneurial spirit have won the day. Straightforward conclusions emerge. There has been an output explosion in the consumer goods segment. This is a win for the Bombay view, as many smaller enterprises, some of them employing surplus rural labor, have set up shop to provide consumer goods. Next, there has been an output explosion in the capital goods segment. This is a win for the Calcutta approach, as many businesses and industries that can beget other businesses and industries have emerged. Then the data indicate that there is a capital-for-labor substitution under way in Indian industry. The importance of this key technological change phenomenon cannot be downplayed. Such a technology diffusion effect is an extremely important industrialization contingency. It is a signal that India could become a manufacturing nation. That outcome will affect the future course of India's late, late industrial revolution.

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The ground realities of behavior by firms and entrepreneurs back up the evidence presented. What conclusions are suggested? India is fertile ground for consumer goods and heavy industry; her firms are globalizing, a major issue not dealt with; and emergent technology entrepreneurship may change the nature of Indian markets. India's industrial growth story is being driven by a variety of entrepreneurs and firms. These are informal-sector entrepreneurs, private enterprises, state-owned enterprises, and multinational corporations. Some Indian firms have the wherewithal to become seriously large global corporations. Many large-scale business expansion announcements are made every day in India. It is the concatenation of such dynamic investment and entrepreneurial activities that will sustain and drive forward India's industrial progress. Because I have framed the story of India's industrial progression as a late, late industrial revolution, it is useful to understand what the original industrial revolutions were all about. The next chapter describes the brief history of industrial revolutions over the course of the last 250 years: what were they, what made them happen, why did they happen where they did, and what were their consequences for various economies, particularly India?

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2 Industrial revolutions

Origins

Economic history includes three groups of events called revolutions. In the eighteenth century, the first industrial revolution occurred. At the end of the nineteenth and beginning of the twentieth century, the introduction of electric power and the internal combustion engine led to a second industrial revolution. Advances in computing, electronics, data storage, and communications networks have led to an information and knowledge revolution. The impact of each revolution has been massive. In many cases, aftershocks continue centuries later. Modern societies have emerged because of the systematic application of scientific knowledge for productive uses. The late Nobel laureate Simon Kuznets had remarked that "The epochal innovation that distinguishes the modern economic epoch is the extended application of science to problems of economic production."[1]

The industrial revolution commences with world leadership in productivity and technology for the past six centuries, initially provided by the region that is the north of Italy and the Flanders region of Belgium. These regions, where the renaissance generally took place after the Middle Ages, played the leading role in the global economy from 1400 to 1600. Then the Netherlands led from 1600 to 1820. After 1820, there were two global technology leaders. The United Kingdom had the highest levels of economic attainment, from 1820 to 1890. Thereafter, the United States took over. In the twentieth century, the leader set the stage for others. It was accepted that the United States was the twentieth-century leader.[2]

Carlo Cipolla had asked the important questions as to why were the Europeans not only able to force their way through to the Spice Islands, but also gain control of all the major sea routes and establish overseas empires? What allowed the Europeans to accomplish the dramatic and sudden transition from a state of insecurity to that of bold

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and aggressive expansion?[3] Why did the "Vasco da Gama era," which had critical effects on India, come into being?[4] The success of Europe during the Renaissance period was based on technological change, and resource substitution effects. Seagoing galleys' oarsmen were substituted by sails. Wind power took over from manpower. Guns substituted for warriors. Gunpowder substituted for swords. These led to the exchange of human energy for inanimate power. By using sailing ships carrying guns, the Europeans accomplished an energy transformation by breaking the bottleneck inherent in human energy use.[5] European maritime expansion was one factor paving the way for the industrial revolution.[6]

The fundamental transformation in human economic circumstances occurred in the late eighteenth century. Around 1780, the steam engine further substituted inanimate for animate sources of power. The steam engine was first used for pumping water out of coal mines. It later replaced water power in driving machinery.[7] A vast new source of energy was opened up. Heat conversion led to the use of new machines, with major consequences of introducing such machinery.[8] Thereafter, new techniques for making iron, by replacing charcoal with coke in iron ore smelting, and a new process for refining pig iron, increased supply and lowered the price of wrought iron. By 1788, there were 59 blast furnaces using coke in the British iron industry. Iron was smelted in blast furnaces with constant blast pressure and rolled mechanically.[9] The products of these mills became the building blocks of the industrial revolution.[10]

The primary consequence of introducing such machinery was the origin of the factory system, the division of labor, and major increases in productivity. By the final decades of the eighteenth century, the British had burst through the labor barrier in textiles.[11] In the textile industry, with which the industrial revolution is associated, by 1820 a worker operating several looms, originally invented by Edmund Cartwright, driven by power, produced 20 times the output of a manual worker. A single railway engine could transport goods requiring several hundred horses. This functionality led to the origin of the term horsepower.[12] These productivity increases, primarily in textiles, in turn stimulated demand for more machines, raw materials, iron, shipping, and communications.[13]

The use of steam-powered engines also removed a binding constraint on human mobility since the invention of wheeled chariots by

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the Sumerians. For over 4,000 years, till the mid eighteenth century, the average speed achieved by wheeled carriages was between four and five miles per hour. Hence, a limited distance was covered in a day. Speed of land travel increased between five and ten times on trains. In sea travel, the steamships could cross the Atlantic in one tenth the time that the fastest sailing ship could achieve. These increases engendered political unification, as well as market coverage, since land that was ten times larger in linear dimensions, and a hundred times larger in area, could be brought within the range of prompt communications from headquarters.[14] The industrial revolution in Britain expanded national wealth and purchasing power. These constantly outweighed the rise in population.[15] Similarly, the use of machines led to average real wages rising between 15 and 25 percent in the years 1818 to 1850, and by 80 percent in the next 50 years. This wages growth led to rising demand for food and other consumer goods, with the steam ships and railways bringing the North American agricultural surplus to be distributed in Britain.[16]

In the mid to late eighteenth century, agriculture and the textile industry formed the basis of both European and non-European economies. In India and China there were many traders, textile producers, and craftsmen. The differences in per capita income were not enormous. Hence, the shares of India and China in global output were substantial. The industrial revolution changed the balance of world economic progress. The British share of total world manufacturing continued to rise, as it became the first industrial nation. As other European nations and the United States adopted the industrialization model, their shares of global manufacturing and per capita incomes rose steadily.[17]

In creating modern industry, inanimate sources of power transcended human biological limitations. The course of an industrial transformation had certain characteristics. These were: the introduction of energy sources via fossil fuels use, the extensive use of mechanically powered machinery, and the widespread use of materials not occurring in nature. These factors interacted to expand output, with the process of output expansion depending on the capture and use of transient energy. The greater the number of mechanical means a system possessed for the recapture, use, and passing around of energy before its discharge from the system, the larger were the cumulative consequences of the energy utilized. Such a process led to the emergence of the scale effect in industry.[18]

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Thus, the British rate of productivity growth in the first half of the nineteenth century became thrice the rate observed for the first half of the eighteenth century. In the second half of the eighteenth century, the rate of growth had been twice what it was in the preceding period.[19] In an industrial economy the role of agriculture diminished, but agricultural productivity increased greatly. Because of industrialization, the structural transformations led to the rise of the mining, manufacturing, and construction sectors.[20] The significant technological improvements involved the use of machinery and mechanical power to perform tasks done slowly by human or animal power, or not done at all. This period in British industry witnessed the application of mechanically powered machinery in the textile industries, James Watt's invention of the steam engine, and the development of the factory system of production.[21]

Why did the industrial revolution happen? The application of science to industry is the distinguishing characteristic of modern industry.[22] Given interactions in human history, there were important and significant relationships between the scientific revolution and the industrial revolution.[23] One of the important historians of the industrial revolution, Thomas Ashton, wrote that the changes were not just industrial, but also social and intellectual. Along with industrial changes, there were commercial, financial, agricultural, and political changes. The intellectual changes were fundamental because they allowed the other changes to occur.[24]

What were these intellectual changes? Key individuals in human history, such as Nicolaus Copernicus, Galileo Galilei, René Descartes, and Isaac Newton, via their scientific achievements, had reinforced ideas in circulation about the practical possibilities of harnessing nature's forces. The science of mechanics, in all its branches, became central to generating industrially usable knowledge.[25] Between the scientific ideas that became established in human knowledge about nature, beginning with Copernicus in 1543, and culminating with Newton in 1687, several other intellectual, ideological, and utilitarian ideas arose within the matrices of knowledge systems.[26] In 1660, Francis Bacon had influenced the founding of the Royal Society for Improving Natural Knowledge in Britain. From then onwards, methods of science such as observation and experiment were applied for utilitarian purposes.[27]

As discoveries occur, the scale of their occurrence expands and changes human consciousness.[28] Within scientific discovery efforts,

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the processes of adapting, assimilating, transforming and general borrowing take place.[29] Intellectual ideas and resources are routinely borrowed.[30] Thus, by 1750 the same scientific knowledge was available in all major languages but its utilization varied substantially.[31] In Britain, prior to the eighteenth century, technology was based on handicrafts, and production had been slow.[32] In the late eighteenth century, the application of scientific knowledge and experimental forms of inquiry into the making of goods, the moving of heavy objects, whether by coal or water, and the creation of new power technologies transformed human productivity.[33] By the second half of the nineteenth century, the flowering of chemical and electrical science theories then provided the foundations for new processes and new industries.[34]

India's role in motivating the industrial revolution

Technology application is an expression of human responses to a changing environment, so that new situations may be met with new ideas, and new actions are generated.[35] What might have been the political economy motivations for the industrial revolution? Every scholar of the industrial revolution, whether Thomas Ashton, Fernand Braudel, Rondo Cameron, Phyllis Deane, Eric Hobsbawm, Paul Kennedy, Geoffrey Owen, Nathan Rosenberg, or Walt Rostow, has agreed that the first industry significantly affected was the cotton textile industry. Eric Hobsbawm had explicitly remarked that: "whoever says industrial revolution says cotton."[36] The first major increases in output growth and productivity were noted in this industry. Between the 1750s and 1830s, the mechanization of spinning in Britain increased productivity in the sector by over 300 times. The cotton industry led industrial change in general, created industrial regions, and led to the development of the factory form of production, an organization type which changed the basis of society forever.[37] The textile industry continued as the mainstay of the industrial revolution till the mid nineteenth century, when the railways took over as the driving force.[38] Profits from the textiles sector financed heavy industrialization and railway building.[39]

The technical problem of mechanization in the cotton industry had been the imbalance between the efficiency of spinning and weaving. The spinning wheel was much less productive than the handloom, speeded up by the flying shuttle (invented by John Kay), and could

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not supply weavers fast enough. Three inventions changed the picture. James Hargreaves' 1764 spinning jenny enabled a single spinner to spin multiple threads at once. Richard Arkwright's 1768 water frame used the idea of spinning by combining rollers and spindles. A fusion of the two machines led to the creation of the mule, developed by Samuel Crompton, to which steam power was applied.[40] James Hargreaves' and Richard Arkwright's innovations established factory production.[41] Subsequently, cotton masters began to build functional factories. By the early nineteenth century, they had lengthened the working day by illuminating their factories with gas. They also bleached and dyed textiles by recent inventions of chemistry, a science which came of age in the late eighteenth century.[42] The impact of the cotton industry extended well beyond its own market environment. While the industry did not directly stimulate the heavy capital goods industries of coal, iron, and steel, it stimulated urbanization as workers clustered in factory towns. This general urbanization provided a stimulus for coal in the eighteenth and early nineteenth centuries.[43]

While the growth of the textile industry slowed down in the middle of the nineteenth century, a textile industrialization age of crisis indirectly led to breakthroughs for coal, steel, and iron, as alternative growth drivers were sought. These were the staple items for railway construction. Between 1830 and 1850, over 6,000 miles of railways were opened in Britain. The two periods, between 1835 and 1837 and between 1845 and 1847, characterized the second phase of the British industrial revolution, when the focus shifted to railways. The railways were largely responsible for the doubling of British iron output between the mid 1830s and the mid 1840s.[44] The railways were revolutionary because they affected ordinary citizens' lives.[45]

The industrial revolution in Britain being the first such revolution, it cannot be explained in terms of imitative behavior leading to the adoption of more advanced techniques, since these techniques were first invented during the revolution, nor to the import of capital or the impact of an already industrialized world economy.[46] Could there be some other reason, along with the interplay of science and technology, which motivated the search for alternatives? India had resounding success as a textile manufacturing nation, in a non-mechanized way, by the seventeenth century. She had made major inroads into British and European markets. A substitution of imports motive proved to be the critical economic trigger for the industrial revolution.[47] Businessmen

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search for ways to reduce costs, as items become more expensive. Technological innovations are sought, to reduce costs. Capital, embodying the latest technology, becomes the important factor of production.[48] Replacement of Indian imports, which created a trade deficit, motivated the search for a technological breakthrough in Britain.

By the year AD 600, Indian cotton had been introduced to Iraq. It then spread to Syria, Cyprus, Sicily, Tunisia, Morocco, Spain, and Egypt.[49] From 1670 onwards, imports of cotton textiles from India had become widespread in Europe. The only pure cotton industry known in Europe at that time was that of India. By the end of the seventeenth century, Indian textiles were a major force in European markets,[50] leading to continuous trade deficits in Britain.[51] Its products, called calicoes,[52] the term originating from the name of the port town Calicut,[53] were imported by the Eastern trading companies. The sales of popular Indian textile items were bitterly opposed by European manufacturers of wool, linen, and silk.[54] In Britain, the Calico Act had been passed in 1701 to ban Indian imports. Its failure led to a second Calico Act in 1721 banning all textile imports.[55] Indian textiles' quality provided a major challenge to European manufacturers, because of the fineness of muslins from Bengal and the brilliant colors and patterns of other cloths.[56] Abbé Raynal remarked in 1777 that "If Saxony and other countries of Europe make up fine China; if Valencia manufactures Pekins superior to those of China; if Switzerland imitates the muslins and worked calicoes of Bengal; if England and France print linens with great elegance; if so many stuffs, formerly unknown in our climates, now employ our best artists, are we not indebted to India for all these advantages?"[57]

In his economic growth theories survey, Walt Rostow included a description from the eighteenth century of the impact on French women of Indian calico textile items: "fruit defendu, les toiles deviennent la passion de toutes les filles d'Eve françaises" ("forbidden fruit, cotton cloth became the passion of every French daughter of Eve").[58] The taste for printed calicoes spread rapidly in Europe, especially in Britain and France. The authorities in France tried various measures, such as confiscation, imprisonment, and fines, but no measures worked. A Paris merchant, based in the rue des Bourdonnais, named Brillon de Jouy, even offered to pay 500 *livres* to anyone willing "to strip in the street any woman wearing Indian fabrics." If this was thought too extreme, he invited citizens to "dress up streetwalkers in

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Indian fabrics."[59] The satirist Daniel Defoe had also mocked the fashion of the day for Indian clothing. In an article in the *Weekly Review*, he wrote:

> Such is the power of a mode as we saw persons of our quality dressed in Indian carpets, which but a few years before their chambermaids would have though too ordinary for them; the chints were advanced from lying upon their floor to their backs, from the footcloths to the petticoats; and even the Queen herself at this time was pleased to appear in China and Japan, I mean China silks and calico. Nor was this all, but it crept into our houses and bedchambers; curtains, cushions, chairs and at last beds themselves were nothing but calicoes and Indian stuffs.[60]

The demand for Indian cotton cloth was so much that it was estimated half the French population wore cotton as opposed to the locally available linen or wool. The French had resorted to duties, and other inhibitions on imports, to protect their domestic industry, but finally in 1757 reduced their tariffs. The quality of Indian goods set off a chain reaction, via a demonstration effect, to reproduce goods of similar quality, at low cost, and enjoy the profits.[61] This provided incentives to develop machinery to produce fine textiles that clumsy European hands could not produce, though hand-weaving methods had long been used in India. A major spur to the industrial revolution in Britain was the motive to substitute Indian imports by domestic manufactures. Richard Arkwright's machines could not spin cloth as fine as Bengal muslin. Samuel Crompton's machine finally achieved the production quality of Bengal hand-woven cloth.[62] There was also a cost motivation, based on an explicit capital for labor substitution,[63] since wages in Europe were six or more times higher than wages in India.[64]

What was the impact on India of British industrialization via the textiles sector? The Indian export trade in textiles had accounted for almost 2 percent of national income in Moghul India.[65] The early nineteenth-century era of cheap textiles from Lancashire effectively destroyed the Indian export trade in textiles, simply because the Indian items were no longer cost competitive.[66] By the first half of the nineteenth century, Lancashire had progressed swiftly, and Britain replaced India as the largest supplier of cotton goods to the rest of the world. The domestic market for yarn and cloth in India then fell into British

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hands.[67] Effectively, for three quarters of a century, the industrial revolution in textiles led to the domination by the Lancashire firms of global and India's domestic trade.[68]

The tables started turning in the late nineteenth century, when India adopted industrial practices and revived Indian cotton production via the use of mechanized spinning and weaving. The modern Indian cotton textile industry was born in 1854, when the first cotton mill powered by steam in Asia, established by Cowasjee N. Davar, became operational in Bombay. Yet, by 1865 there were only ten cotton mills. Expansion occurred in the 1870s, with 47 cotton mills in operation by 1875, and 79 cotton mills by 1883.[69] By then, however, mercantile exploitation of the Indian economy had been under way for over a century by the British. First, between 1757 and 1813, when it lost its trading monopoly, the East India Company used the revenues it collected on behalf of the Moghuls earlier, and then themselves, to buy goods in India for sale in England. In other words, the revenues of India provided the East India Company with capital for their trading operations. These revenues were euphemistically called the "investment" of the Company in India.[70]

After 1813, and the loss of the East India Company's trading monopoly, the industrial revolution generated a new economic relationship between India and Britain. Between 1813 and 1858, when the governance of India passed from the hands of the Company to the Crown, the pattern of activity changed to one of free trade whereby India became the market for Lancashire products,[71] while she supplied the raw materials to Britain for processing.[72] Arnold Pacey wrote:

> After 1803, however, the arrogance of conquest was reinforced by the rapid development of British industry. This means that Indian techniques which a few years ago earlier seemed remarkable could now be equaled at much lower cost by British factories. India was then made to appear rather primitive, and the idea grew that its proper role was to provide raw materials for western industry, including raw cotton and indigo dye, and to function as a market for British goods.[73]

After 1813, imports of cotton fabrics into India rose from a million yards in 1814, to 51 million yards in 1830, and to 995 million yards by 1870. This drove out the traditional domestic producers and in the process led to India's de-industrialization.[74]

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Britain and the industrial revolution

Why in Britain? Why not elsewhere? From the middle of the seventeenth century, British science had come wrapped in an ideology encouraging material and commercial gain.[75] In Britain, many networks linked scientists with businessmen.[76] Scientists were heroes, with exploits popularized in newspapers and journals. Science became a belief system yielding power over nature. Initially in Britain, and then gradually throughout Western Europe, the population learned about science in schools and lecture halls, and picked up its contents from general textbooks. In Britain, science integrated differently with the social environment. By 1750, people approached the productive process through an implicit mechanical lens by visualizing it as something to be mastered by machines.[77] On an abstract level, the process was conceptualized in terms of weights, motion, the principles of force and inertia. Copernicus had argued that the sun lay in the center of the universe. In the next generation, Johannes Kepler established the laws of planetary motion, while Galileo had discovered the keys to the local motion of earth-bound bodies. In the 1660s, Robert Boyle perfected an air pump that convincingly displayed the presence of vacuum, and discovered the laws of gases. Isaac Newton demonstrated the law of universal gravitation, thereby proving the importance of Kepler's planetary laws and elaborating on the mechanics of Galileo.[78] One can now easily see how tying together the ideas of Galileo, Kepler, Newton, and Boyle would have led to the subsequent invention of the steam engine.

Next, take the case of Jean Desagulier, a Huguenot refugee in Britain who was a lecturer and official experimenter for the Royal Society. He had authored a book in 1744 on mechanical knowledge titled *A Course of Experimental Philosophy*. This text, while discussing the steam engine, contains the first instance when anyone writing, in any language, spelled out the critical insight that mechanization undertaken by engineers could enhance the profit of entrepreneurs by reducing labor costs. By the middle of the eighteenth century, a materialist ideology of commercial development was linked in the minds of entrepreneurs with mechanical applications, and writings of people such as Desagulier appealed directly to them. British science, in the form of Newtonian mechanics, it is suggested, directly fostered industrialization.[79] Margaret Jacob states that "there was one common element

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that would prove immensely germane to the mechanization of the manufacturing and transportation systems: the belief that science and technology could control nature and that creativity in both was desirable."[80]

While workers could be brutalized, when viewed in these terms, manufacturing costs were reduced by the use of machinery instead of people, thus increasing profits.[81] By the end of the eighteenth century, capital and ingenuity had promoted industrial progress for reasons that had much to do with scientific and commercial cultures.[82] By 1800, a British mechanical vision had synthesized nature with a moral economy of prosperity. The scientific innovations that abounded were applied to practical matters by scientists.[83] The pervasive new scientific learning fueled British entrepreneurs' imaginations as to how to make profits.[84] It led to the creation of the technology entrepreneurs' class.[85] Industrial entrepreneurs either possessed technical skills, or hired people who did. They needed to assimilate applied scientific knowledge along with business skills.[86]

Sharing a common vocabulary of technology, engineers and entrepreneurs, such as Mathew Boulton and James Watt, could talk the same mechanical talk, were able to visualize the physical world in its mechanical and commercial terms, and align their common interests into partnerships.[87] The impact of this alignment had permanent effects on the evolution of the world's economy. In Britain itself, the advance of the industrial revolution by 1820 had attracted attention. Economists even formulated a law, known as Say's Law, named after Jean-Baptiste Say, which stated that supply created its own demand. A useful modern statement on Say's Law is by the historian Fernand Braudel, who said:

> Another, more logical way to put the same thing is that the production of any good which will sooner or later be in supply on the market, has already led, in the very process, to a distribution of money: the raw materials have had to be paid for, the costs of transport found, and the workers given their wages. Once this money has been distributed, its normal function is to reappear, sooner or later, in the form of a demand or if one prefers, a purchase. Supply makes an appointment with itself.[88]

The economic impact of the industrial revolution is best assessed by reviewing patterns of British industrial growth in the nineteenth century. In this period, Britain experienced relatively high industrial

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growth. The growth rates touched almost 48 percent, over the decade between 1820 and 1830. A slowdown took it to 37 percent over the decade between 1830 and 1840, and to 39 percent over the decade between 1840 and 1850. Thereafter, it slowed down considerably.[89] But the manufacturing revolution had triggered Say's Law!

The American experience with industrialization

The American experience has been fundamental to global industrialization. The phenomenon of growth in cotton, electricity, and cars is history. America, however, is the home of mass production of machinery. Advances in mass production, particularly of the Colt revolver, the Winchester rifle, the clock, and the sewing machine, and the use of the assembly line, initially in Chicago and Cincinnati slaughterhouses, and then in mass producing cars, originated in America.[90] Yet, America's industrial revolution, along with that of Germany, came after that of Britain. What were its contours? What lessons can be learnt from these experiences? Industrialization in the United States and in Germany was characterized by scale and scope economies. These were based on unique designs for production and distribution organizations. Nevertheless, these innovations came much later, in the latter half of the nineteenth century, when Britain had been an industrial power for a century.

What was the American position when the first industrial revolution was under way in Britain in the late eighteenth century? The United States did not have a great reputation for scientific originality, because many of the fundamental discoveries had originated in Europe in the fifteenth and sixteenth centuries. At the time of the American Revolution, the American economic position was so backward that it was felt that the fledgling republic could not play a role in the affairs of nations. Enthusiastic imitation had to be the norm. The Americans had to beg, borrow, and steal ideas in order to commence industrialization, but inventors and innovators such as Eli Whitney and Thomas Edison transformed America and the world.[91]

In 1790, America had a population of 4 million compared to Britain's 14 million people, the economy was based on farming, and just about 6 percent of the population lived in urban areas. Also, one in six persons was a slave.[92] By the late 1790s, Britain was supplying

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80 percent of American imports while taking half its exports.[93] The money system was chaotic, with promissory notes or bills of exchange being more valuable. There were no decent circulating mediums of exchange. In 1789, in the provisional state of Tennessee, it was proposed that the Governor be paid 1,000 deerskins annually; the Chief Justice and the Attorney General were to receive 500 deerskins, the County Clerk was to receive 300 beaver skins a year, the members of the General Assembly were to receive a daily allowance of three raccoon skins and a justice of the peace could charge one muskrat skin for signing a warrant.[94]

In a backward economic and social milieu, the importance of Eli Whitney's invention is out of all proportion to subsequent inventions that emerged from the United States. Eli Whitney was the son of a small New England manufacturer who made nails, and had been sent to Yale. Finding little employment in the region, after graduation, Eli Whitney went to Georgia as a private tutor. Finding conditions unwelcoming, he stayed on briefly in South Carolina where he happened to notice the large amounts of labor required to harvest rice, then a staple item in American trade. In social gatherings, he had heard that if a machine could be found to clean cotton, which was the other staple export item, it would be of great consequence. While the cotton being grown in the Caribbean to supply Britain was the long staple kind, the type grown in the American South was the short staple variety. Since tobacco and rice were the key trade items, there could be major impact of developing the cotton sector; however, cleaning the short staple variety cotton was difficult.

Eli Whitney's development of a simple cotton harvesting and cleaning machine, called the cotton gin, in 1793 revolutionized American cotton growing. In 1791, cotton exports were 189,000 pounds, but fell to 138,000 pounds a year later. By 1793, it had grown to 487,000 pounds, to 1,601,000 pounds in 1794 and to 6,276,000 pounds in 1795. By 1800, about 18 million pounds of cotton were being exported from America.[95] This was the biggest breakthrough in production scale that the world has ever seen to occur in less than a decade. Cotton at that time was the basis for the world's largest industry, textiles. The United States played an increasingly vital role in the British industrial revolution by providing the raw cotton, enabling Britain to become the world's loom. The United States eventually supplied Britain with 75 percent of its cotton. By 1860, Britain consumed half the world's

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cotton supply. It had 3,000 factories, using 400,000 powerlooms and employed 500,000 people in cotton textiles.[96]

The simplicity of Eli Whitney's gin was in its original use of standardized parts. This made it possible to rapidly mass manufacture the machines and diffuse them through the relevant parts of the country. Eli Whitney is one of the founders of the mass production approach; one of those, according to Jonathan Hughes, of "American genius that put the system of standardized output and interchangeable parts to work for the good of mankind."[97] This production approach soon had impact elsewhere. The spillover effect was that many British firms in the metalworking industries adopted the American system of manufacture by making standard machinery products from interchangeable components.[98] In the 1850s Britain was the industrial leader; by the end of the nineteenth century the United States had overtaken Britain. This facet of relative performance is also apparent from the slowdown in British industrial growth in the latter part of nineteenth century. By the final decades of the nineteenth century, British industrial growth rates were between 1.7 and 1.8 percent per annum.[99]

The growth of American manufacturing after the Eli Whitney era led to a change in economic processes. This was the substitution of capital for labor in the American economy. Differences in resource endowments, and demand conditions, materially affected the economic development of a country. They influenced the type of product characteristics and manufacturing processes to be used.[100] An abundance of natural resources meant it was rational to use approaches intensive in natural resources consumption, rather than less freely available labor and capital equipment. Thus, America became an early pioneer in the development of machinery for woodworking, because of an abundant supply of resources from the forests.[101]

In America, farmland was plentiful and bountiful. This attracted people to work on the land as farmers. Labor, especially trained labor for manufacturing units, was in very short supply. America was a dynamic society built on settlers. It was one in which labor was more mobile relative to Britain. Opportunities for self-employment were greater. Restrictions on personal freedom, which factory apprenticeships entailed, were unacceptable. Wages were high, so as to retain men in manufacturing plants. High wages, however, meant uncompetitive products. Thus, substitution of capital for labor enhanced the capital-to-labor ratio in firms.[102] Firms replaced a more expensive factor of

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production by a cheaper factor. As labor became more expensive, it was replaced by capital, which embodied the current technology in vogue, as a factor of production.[103]

Firms combined the quantities of labor and capital which maximized profits, the motivation being to maximize the cost reductions by substituting capital and technology for labor where feasible. Therefore, technology substituted for labor inputs. Consequently, American manufacturers looked for manufacturing methods that economized on the use of craft skills, as principally embodied in human beings, which was the expensive labor factor of production, and used greater proportions of machinery. The manufacturing processes to do so involved greater standardization.[104] At the Crystal Palace exhibition in Hyde Park, London, in 1851, the British coined the phrase "American system of manufactures" because of the distinctiveness of American products. These referred to goods which were produced by specialized machines, were standardized, and used interchangeable parts.[105] This standardization approach had evolved in United States government armories, such as the Springfield Arsenal in Massachusetts, for the production of small arms.[106] It was then taken up by the manufacturers of items such as bicycles, sewing machines, typewriters, and other products for which there was a large and homogenous demand.[107]

Behind standardization was the prior development of a capital goods sector. In the early economic history of the United States, the capital goods sector primarily dealt with metal forming and metal shaping. There were learning centers where metal-working skills were developed, and then transferred to the production of a sequence of standardized products. The commonality of production processes was diffused across a range of industries, so that there was mutual learning across industries. Production processes for items such as firearms, clocks and watches, agricultural machinery, hardware, sewing machines, typewriters, office machinery, bicycles, and automobiles had similarities and common features.[108] Such similarities engendered knowledge spillovers. These steps would lead to the generation of manufacturing efficiencies across the production systems for the entire range of items required for daily human activities.

On October 21, 1879, Thomas Edison took a piece of cotton thread, put it in a mold and carbonized it in a furnace. He had been experimenting for some time to develop a long-lasting filament for an incandescent lamp. He put the carbonized filament in a glass container,

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removed the air in the container and passed electricity generated from a small dynamo into the container. The filament lit up and stayed lit for over 12 hours.[109] Thus, the electric light bulb was born, and soon thereafter the electric power industry. Thomas Edison had been trying for some time to develop a working light bulb but his previous attempts to make the filament stay alight for any length of time had failed.

The broad principles behind electricity and electric lighting were already known. In 1801, Humphrey Davy had demonstrated carbon arc lighting, in 1831 Michael Faraday had invented the dynamo to convert mechanical power to electric power, and by the 1860s Joseph Swan held a British patent for incandescent lighting.[110] Thomas Edison's breakthrough in 1879 had come because he had visualized the overall system of incandescent lighting incorporating the glass container with a vacuum and the filament to which an electric current had to be applied. The broader industrial system involved wiring, switches, meters, a power plant, and light bulbs, so that homes, factories, businesses, and other users of electricity could utilize it for lighting and other purposes. It was this same system-visualization which led to the modern power industry. By the end of the 1880s, Thomas Edison had laid the foundations of Consolidated Edison and General Electric, two corporations which are still major economic influences today.[111]

The growth of the electrical industry in the United States rested on organizational skills involving the construction of an integrated system for generating, transmitting, and distributing electricity and the recognition that once the system was in place there was a mass market waiting to be tapped.[112] The new electrical energy source not only transformed the mechanical processes of production within factories and created a new form of urban transportation, but also revolutionized the making of many metals and chemicals.[113] The economic consequences of electrification were high. The average cost of each lamp had fallen from 70 cents in 1881 to 31 cents in 1884. By 1888, there were 185 electricity stations in the center of towns supplying 386,000 lamps, and 1,281 dispersed stations supplying 344,000 lamps. The United States Census of 1880 had listed 40 electrical establishments employing fewer than 900 workers. By 1890, there were 189 electrical establishments listed employing 46,000 workers. Power stations in 1890 employed 2,000 staff, and by 1902 there were 3,620 power stations employing 30,000 staff.[114]

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No less compelling than the achievements of Eli Whitney and Thomas Edison were the achievements of Henry Ford. His greatest triumph, and that of the American system, was the mass-produced automobile. American entrepreneurs were good at taking inventions made elsewhere, like the internal combustion engine, and adapting them to local conditions.[115] As innovators, they could put together separate components that others had originated and devise a system that would transform economic life. Henry Ford changed the American economy by making the automobile ubiquitous.[116] The automobile turned the static mechanical energy available in a stationary engine dynamic by putting wheels on that engine. It made human beings more productive by replacing the energy of several horses. It made transportation more efficient. The multiplier effect of such efficiencies on national output was massive and inestimable.

The automobile (the term includes cars, trucks, and buses) changed American topography. It transformed the landscape by creating suburbs. Cities in America spread out in all directions. They still continue to do so. In many parts of the country, the ratio of automobiles to people is greater than 100 percent. It is believed that the area of southern California designated the Greater Los Angeles Area contains many more automobiles than people. In my town of current temporary residence, Dallas, the entire metropolitan region covers an area 80 miles in distance in its north–south axis and 90 miles in distance in its east–west axis. The area has more motorway miles than contemporary Great Britain. In this area, there are more automobiles than people. The sociological impact of such changes was substantial. New shopping centers, malls, and an entire urban culture based on the automobile were created. This was in diametric opposition to the urban street culture of Europe, where street life and a café society were the norm in cities. The automobile revolutionized the transport of goods. It created a whole multi-billion dollar ecosystem of industries devoted to its production and maintenance.

A bigger impact was on human welfare. The automobile offered particularity. Anyone owning a car could decide when to start and finish a journey, and where that journey was to be. The route from place A to place B could be self-chosen, subject to available roads. The journey could be made alone or with others. The choice of speed, subject to institutional constraints, was discretionary. An individual would no longer be isolated if one possessed a car and could set out

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on a journey.[117] The automobile expanded human choice and freedom in critical welfare-enhancing ways that have yet to reach their limits. The ubiquity of the automobile was apparent in the United States by the start of the First World War. It was only in the 1950s, after the Second World War, that European countries accepted personal cars as a necessity. It is in the first decade of the twenty-first century that the automobile is coming into its own in China and India.

Where does Henry Ford fit? Apparently, the years of his greatest success, between 1903 and 1926, were the years in which the national income of the United States grew at 10 percent per annum, but that is a coincidence. Henry Ford was not an inventor, but his ability to translate his vision of making what was then a rich man's toy available for everyone had an influence on economy and society beyond what the statistical correlations imply. The instrument Henry Ford used, the Ford Model T, made the automobile an item of mass production and consumption. His methods for attaining mass production have become a legend. The Model T was designed as a light and strong but also a powerful and efficient automobile. Most of all, it was cheap. Originally planned to be priced at $850, it was sold only in black color. Henry Ford also decided to manufacture just one model of car.[118] These simple strategies revolutionized production economics. Anyone could own a car, and be on equal terms with other car owners. The Model T made automobiles classless. It commenced the market democratization era by democratizing transportation.

What would further revolutionize the manufacturing sector for ever was the introduction of the assembly line in the Model T's manufacturing process. The work moved to the worker. The large River Rouge plant in Detroit was fully vertically integrated. Ore would come in at one end, be processed into metal, have components bought from outside added, and then emerge as a finished car. This approach may upset make-or-buy purists who will wish to examine the marginal costs and benefits of integration. But at that time this approach worked wonders. In conjunction with fundamental redesign of the materials handling process, such that a tributary of parts flowed into the river that was the assembly line,[119] the process of assembly line manufacturing enhanced productivity substantially, in conjunction with Henry Ford's $5 per day wages strategy of 1914, which he had called "efficiency engineering."[120] The assembly line was a successful innovation that was copied in all industries and all countries wherever feasible.

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The impact of the assembly line can be measured by output. In 1909, the first year, 10,600 Model T cars were made. By 1926, when the model was withdrawn, 15 million Model T cars had been made. By the time the scale efficiencies had percolated through the system, the costs had dropped rapidly. The most basic utility version of the Model T could be purchased for $250.[121] At peak production, one Model T came off the assembly line every 15 seconds, although usually it was one every 45 seconds. Till 1926, half the cars made and sold in the United States were Fords. The output increase of over 150 times was due to the introduction of the assembly line. This took scale economies' exploitation to their logical conclusion.[122]

Eli Whitney had belonged to the first industrial revolution era. Thomas Edison and Henry Ford are the originators of mass manufacturing and production approaches that matured during the second industrial revolution. Their lessons became widespread. In many industries, expansion of output could occur by adding more machines and workers, but would not increase costs linearly.[123] Now costs would rise incrementally. In the 1880s and 1890s, new technologies of mass production brought about the second industrial revolution. This further enhanced capital–labor substitution in the American economy. In the process, America also commenced the steps towards becoming a knowledge economy.

The democratization of invention

The interplay of scientific and technological thoughts led to the initial launch of the industrial revolution. Its commercial propagation, initially via the cotton textile industry, had economic motivations as drivers. The potential textiles market to be accessed was large. India had earlier shown that it could be captured. While these intellectual and economic processes were at play, institutional changes promoted the use of technological innovations to further humanity's material progress.[124] These institutional changes have been described as important non-economic factors that led Europe to escape from poverty.[125] The first is the role of government policies in the late eighteenth and nineteenth centuries. The first set of such policies related to the decentralization and secularization of commerce. By the middle of the nineteenth century, the United States, and several other European countries had given enterprises rights that were a grant of authority to make a

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number of decisions that earlier were made by political or religious authorities.[126]

The second set of policies led to economic freedom to undertake commercial activities free from political interference. The relaxation of political control over the commercial and economic sphere took several forms. Individuals were authorized to form enterprises with few political restrictions. These enterprises could acquire goods and hold them for resale at a profit, again with hardly any restrictions. Then enterprises were authorized to add activities and to switch from one line of activity to another that seemed more promising with no restrictions.[127] The third set of policies led to property rights' inalienability. While the assets of a business enterprise, and its profits as accumulated from its activities, could be taxed, its property came to be regarded as immune from arbitrary seizure or expropriation by political authorities.[128] The placing of systematic limits on the exactions of the government increased commercial security and gave protection to entrepreneurs' expectations. Contractual arrangements began to be enforceable by courts. These replaced a "might is right" syndrome, whereby force or status could be used as the enforcement device. Hence, uncertainties as to what dispute resolution mechanisms might yield were reduced.[129]

The property rights innovations led to the democratization of invention. The emergence of the patent system, by 1700, to protect private property in knowledge, has been given considerable prominence as an institutional factor that set the stage for the industrial revolution in Britain.[130] Subsequent analyses have borne this out. An evaluation of the connection between the patent system and inventive activities during the British industrial revolution found that the British patent system had stimulated inventors' efforts. In addition, a class of professional inventors, who would generate patents to gain financially, via their sale or licensing, arose. While small in size in that period, patenting activities led to the development of technology markets.[131]

Similar phenomena were taking place across the Atlantic. The framers of the United States Constitution were familiar with British precedents and provided for economic inducements to be enshrined in the Constitution. Accordingly, an intellectual property clause, providing for the patent and copyright systems, appears in the very first Article of the Constitution, whereby Congress was instructed to "promote the Progress of Science and useful Arts, by securing for limited Times to

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Authors and Inventors the exclusive Right to their respective Writings and Discoveries."[132] A result of such efforts was the world's first modern patent institution with low costs of filing and relatively rapid development of mechanisms for enforcement. These factors encouraged the Americans to be enthusiastic about establishing claims to intellectual property. By 1810, the United States had far surpassed Britain in per capita patenting.[133] The United States patent system was cheap, with costs being 5 percent of patenting costs in Britain. It included an examination system, by technical examiners, facilitating the use of patents as general assets to be sold, licensed, or given as loan collateral. This feature became important, since technologically creative individuals without money to exploit inventions directly could still engage in discoveries.[134]

These developments were central to patenting growth, as they motivated individuals with relatively modest amounts of skills and capital to invent. This led to the rapid growth in inventive activity in the United States during the early nineteenth century, which coincided with a substantial broadening of the segments of the population participating in the process of patenting. The United States patent system extended property rights in technology to a wide section of the population. Invention became a democratic activity in the United States throughout the century. Though individuals who had studied at institutions of higher learning were among many well-known inventors, the growth in patenting during the beginning of American industrialization was marked by a disproportionate increase in invention by ordinary citizens, sometimes with little formal schooling, operating with common skills and knowledge, rather than by elite with specialized and technical expertise or extensive financial resources.[135]

This process of democratization of invention was grounded in the access to economic opportunities available in an environment where enterprises operated on a small-scale, markets were rapidly expanding, and the barriers to entry were modest. An extension of property rights in technology to the general population meant that persons of humble origins were, by design, well represented among the inventors of the United States.[136] These circumstances made it easier for the United States economy to realize major increases in invention and productivity growth, once conditions raised the private returns to inventive activity and encouraged an allocation of individual resources in that direction.

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The democratization of markets

Important drivers of American productivity were scale and scope economies. These arose from organizational innovations. In several industries, expansion of output came about because of a critical change in the ratio of capital to labor. This would occur by improving and rearranging inputs, by using new or improved machinery, by reorienting production processes, by integrating several intermediary processes required to make a product within a single facility, and by increasing the application of energy.[137] Hence, there was a systemic reorganization of production toward commonalities and volume. In capital intensive industries, new processes of production were invented or existing ones improved.[138] In the capital goods sector, the functionalities were utilized for the manufacturing of complex light, standardized machinery items through the fabrication and assembly of interchangeable parts, and for the production of industrial machinery and chemicals by a series of interrelated mechanical and chemical processes. In many such high capital-using industries, investments in new facilities further increased the ratio of capital to labor. Production units achieved greater economies of scale. The cost per unit dropped more quickly as the volume of materials being processed increased.

These cost advantages, however, could not be fully realized unless a continuous flow of materials through the plant ensured capacity utilization. Thus, two decisive factors in determining scale and volume economies were the rated volumetric capacity of the plant, and the throughput, which was the amount actually processed within a specified time period. In industries using large amounts of capital, the throughput needed to maintain minimum efficient scale required careful attention not only to flows through the processes of production, but also to the flows of inputs from suppliers and the flows of outputs to intermediaries and final users. Hence, along with scale economy considerations, materials acquisition and handling assumed importance.[139] Because of scale and scope economies, at the end of the nineteenth century a petroleum producer could reorganize the processes of production by shutting down some refineries, reshaping others, building new ones, and coordinating the flow of materials through refineries and to consumers.[140]

The increasing scale of manufacturing operations also brought with it the need for well-trained and technically competent managers.[141]

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As production volumes rose, and expensive investments were made in high output machinery, informal arrangements were inadequate. Tighter discipline and closer supervision were needed. In the 1890s, Frederick Winslow Taylor, a production engineer at a leading steel company, was brought in to work out new ways of organizing and motivating shop-floor employees in order to ensure that machinery was used productively.

Simultaneously, in the second industrial revolution, the Germans began to exploit economies of scope. The large German chemical plants produced hundreds of dyes and pharmaceuticals from the same set of raw materials, and the same set of intermediate chemical compounds. Bayer, Hoechst, and BASF are still famous names today. They were the initial companies to exploit the cost advantages of scale and then scope, and reduced the per-kilogram price of a new synthetic dye by about 90 percent between 1869 and 1886. A new dye or pharmaceutical added little to the production cost, and product additions permitted a reduction in the unit cost of other products.[142]

The second industrial revolution led to the origins of industries such as tobacco processing, grain processing, and the production of alcoholic beverages, sugar, vegetable oil, and other foods. It transformed oil refining and the manufacturing of steel, copper, aluminum, glass, abrasives, and other materials. It created new chemical industries that produced synthetic dyes, synthetic fibers, and fertilizers. It led to the development of light machinery for sewing and agricultural and office uses, and to the development of heavier machinery items such as elevators and refrigerating units.[143] The use of manufacturing practices,[144] such as the mass production method of fabricating and assembling fully interchangeable parts, adopted during the second industrial revolution, appropriate business organizations,[145] and the scale and scope economies that these practices engendered, had a significant impact on the world economy in the twentieth century. These practices directly led to giant-scale production. They shaped consumer demand, eventually leading to the creation of markets in novel products. They created new activities, such as home sewing, and changed the nature of consumption.[146]

What did manufacturing economies of scale and scope do to market structure? The use of telegraphs helped integrate markets and bring about a change in activity coordination. Prior to the American Civil War, general merchants had been the important determinants of

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market demand, as the principal United States economic activity was trading. Merchants aggregated demand. Merchants were general traders with a diverse range of activities. Demand satisfaction was based on craft production. As a result, specialization in manufacturing was low and inefficiencies high. Manufacturing economies of scale and scope led to production specialization. Then, manufacturing economies of scale and scope led to distribution economies of scale and scope, and the development of professional management because of the need to coordinate different aspects of the value chain. In sum, the manufacturing and distribution economies of scale and scope led to the integration of markets, expansion of demand, a retail revolution, and the origins of professional management.

By the late nineteenth century, the first and second industrial revolutions had changed the transportation sector. By the 1870s, in the United States particularly, railways, telegraph, steamship, and cable systems operated as a cohesive transportation and communication network. Because of integrated communications, materials could flow in and finished goods be moved out at high speed and in high volume. These led to major changes in distribution practices. A review of the development of railways in the nineteenth century is useful. By the middle of the nineteenth century, when the United States started building railways, Britain was at its peak of railway construction. Thereafter, the growth of railways in the United States, elsewhere in Europe, and in the Americas, particularly in Canada and Argentina, was substantial. These railway developments had a major impact on the nature of world trade, as farmers from the North American prairies, the South American pampas, and the south Russian steppes starting exporting agricultural items to the world.[147]

On the mass consumer products side, again in the United States, new volume wholesalers and new mass retailers evolved. These were department stores, catalog and mail-order stores, and chain stores. The numerous and widespread location of the stores meant that goods were available widely. If a customer could not go to buy the goods, they would come to her via mail.[148] Wholesalers and retailers came into being after the transportation and communication networks made possible high-volume and high-speed shipments.[149] The merchandisers, wholesalers, and retailers also exploited economies of scale and scope generated by the second industrial revolution. The core activity of a volume distributor was buying. The coordination of the flow of

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purchased items through facilities became important, and the turnover of goods processed was to mass distributors what throughput had been to those engaged in mass production. The greater this volume, the lower the unit costs were.[150] Low unit costs of items, their widespread availability, and the retail revolution engendered the democratization of markets.

This process of market democratization first occurred in the United States. By 1913, it was the largest economy in the world. It produced over 33 percent of the world's industrial output, which was just under the combined total output for Germany, Great Britain, and France put together. In 1929, the United States produced over 42 percent of the total world output.[151] The United States steel production rose by 25 percent between 1913 and 1920, while steel production in the rest of the world fell by 33 percent.[152] In general, the industrial revolution had been accompanied by a consumer revolution, as there was a large and rapid increase in the consumption of consumer goods. Such consumer revolutions had occurred in Britain and Europe by the second half of the eighteenth and the nineteenth centuries.[153] The scale of the American industrial revolution engendered such an extensive low-cost democratization of markets that a consumer revolution became axiomatic.

In the twentieth century, after the Second World War, substantial industrial growth occurred in the developed countries. For a generation, these countries accounted for 75 percent of global production and 80 percent of manufacturing exports. There were also smaller industrial revolutions elsewhere in Europe, such as in Spain and Finland. Agrarian countries such as Bulgaria and Romania acquired large industrial sectors. In the developing world, many countries were newly industrialized.[154] Luxury items became necessities. Take the refrigerator, the home washing machine, and the telephone. In 1971, there were 270 million telephones in the world, mostly in North America and Western Europe.[155] Today, there are 2 billion telephone connections, mostly in China and India. The prime driver of this phenomenon has been wireless technology, its widespread diffusion, and the capital intensity of processes.[156] Such capital intensities are a surrogate for knowledge embodiment.

Scientific research has found practical applications. These applications have transformed the activities of daily living, not just in rich countries but also in poor ones. Good Indian examples are the impact

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of the radio, the television, the green revolution in agriculture, and now mobile phones. The operations of Engel's law, by which the poorer spent a greater proportion of their income on basic necessities, has been shrinking, as people have more to spend on other items that matter.[157] By engaging the largest possible number of human beings in enjoying the fruits of technology-led growth, the industrial revolutions have led to the democratization of markets.

India's non-participation and its consequences

India was an important industrial power at the beginning of the industrial revolution.[158] At that time, the primary global industry was cotton textiles. In that sector, India was the world leader. She is not now one in any sector. India did not participate in the industrial revolutions. Why she did not participate is a big question. One can take a simplistic but system-oriented view that suggests a lack of overall agricultural development led to the continuation of rural poverty, and the lack of a robust market with adequate purchasing power. Economic conditions were not conducive. Additionally, there was inadequate human capital development through lack of scientific training and education. The basic human capital skills to permit participation in an industrial revolution, driven by science and technology, were lacking. Even if finances were forthcoming, the institutional structure supportive, and entrepreneurs willing to take the risks, which they might not have had inclinations for, there was inadequate human capital to support involvement.

The consequences of industrial revolutions are the democratization of invention and then, as some of these inventions succeed, the democratization of markets. The impact on national incomes is the measure of these effects of an industrial revolution. Indian industrialization seriously started only after independence in 1947. The late Nobel laureate Octavio Paz had written that "A recurrent theme of Indian history is the clash of civilizations."[159] The late C. S. Venkatachar expressed India's historical and psychological dilemma thus: "In the days of her nationalism India rejected the idea of westernization under an alien agency. After independence she decided against the continuance of traditional forms of her society and the closed cultural circuit which it implied, by opting for vigorous modernization."[160] Modernization of the economy and society, along successful Western lines, started only after independence.

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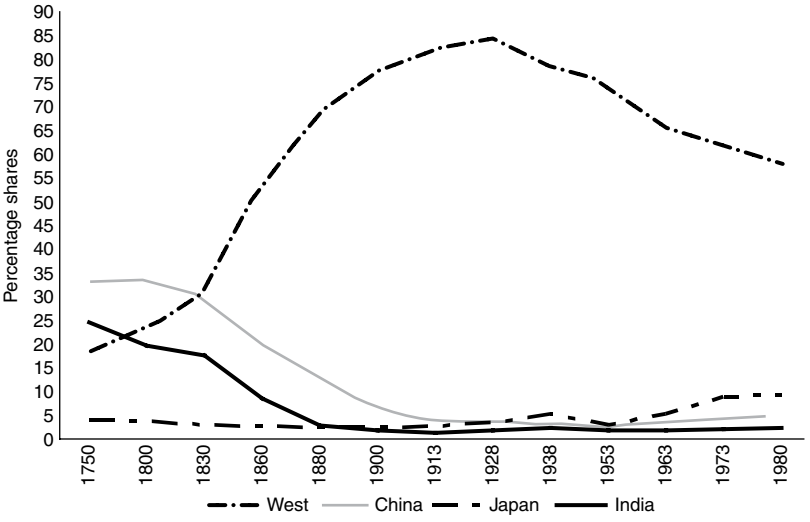

Figure 2.1. Share of manufacturing in the world economy of various countries from 1750 to 1980.

Source: Table A5.

Paul Bairoch[161] and Paul Kennedy[162] documented the changing shares of countries in world manufacturing output. I have provided details from Paul Kennedy's work in another chapter. The details of Paul Bairoch's numbers are provided in Table A5. In this chapter, based on the documentation by Bairoch, I discuss the changing shares of Western economies, China, Japan, and India in world manufacturing output from 1750 to 1850. Some countries' details are displayed in figure 2.1. The figure shows that both China and India slipped and fell from pole positions, after 1750 and till 1980. Since then, China has picked herself up. What is apparent is how the two industrial revolutions changed the relative position of the West by the middle of the twentieth century, how engaging in such a process changed the fortunes of Japan, after engaging in the process, especially during and after the Second World War, assumed a leading position in global manufacturing.

What did not participating in the industrial revolution mean for India? I used data that were generated by the late Angus Maddison to address this issue. The data generated relate to per capita income

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Table 2.1. ***Relative per capita incomes from 1820 to 1989 with India's income in 1820 as the base case benchmark***

	1820	1890	1950	1989
Germany	184	339	672	2,807
India	100	106	102	223
Japan	124	172	331	3,130
United Kingdom	296	690	1,153	2,759
United States	249	633	1,756	3,731
Base case benchmark: India in 1820 = 100				

Source: Calculated from the data in Angus Maddison, Explaining the Economic Performance of Nations, in W. J. Baumol, R. R. Nelson and E. N. Wolff (eds.), *Convergence of Productivity: Cross-National Studies and Historical Evidence* (New York: Oxford University Press, 1994), Table 2–1, 22–23.

levels for a variety of countries from 1820, when the first industrial revolution was on its way, till 1989.[163] I take a sample of the countries that were used, and convert all per capita income levels to a benchmark case of Indian income in 1820. The use of this period permits an assessment of where India was in 1820, and where she had come by 1989. By then, the first and second industrial revolutions had their impact. The year 1989 also precedes the appearance of South Korea and, latterly, China as major industrial countries. It precedes the information and communications technology revolution, in so far as it has had an impact on India. The indices computed are given in Table 2.1. The countries compared are Germany, Japan, the United Kingdom, and the United States. Germany is a large manufacturing economy, and participated in the second industrial revolution. Japan was a late starter, but has achieved miracles; the United Kingdom has had a very long relationship with India; and the United States is the twentieth-century exemplar.

In 1820, India's per capita income, while the lowest because of the size of its population, was large in absolute terms relative to the other countries. By 1820, the first industrial revolution had kicked in and the United Kingdom's per capita income figure was 296 relative to India's figure of 100. Japan, however, was a small and feudal agrarian economy, and its per capita income figure was 124. By 1820, the cotton boom had occurred in the United States, and its

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per capita income figure was 249, but on an extremely small population base. The industrial revolution changed everything. By 1890, India's per capita income had reached an index value of 106, while the United Kingdom and United States per capita incomes were 690 and 633 respectively. Japan was still just commencing its own industrialization, but it had reached a per capita income index value of 172 in 1890.

In 1950, after the world wars and independence, India's per capita income index value had taken a drop to 102, relative to 1890. It was just 2 percent more than in 1820.[164] By then, Germany, having industrialized in the late nineteenth century, had charged ahead, the United Kingdom income index value was ten times that of India, and the United States was the leader of the pack. In 1989, two years before the fundamental reforms in India were implemented, India's per capita income index was 223. It had more than doubled since she became a republic in 1950. By then, the others' incomes had leapt ahead. Germany and the United Kingdom were relative laggards compared to the United States and Japan. They too displayed per capita incomes that were at least ten times larger than India's in 1989, and over 20 times larger than India's in 1820.

The important countries to be compared with are Japan and the United States. Let me take the United States first. Like India, the United States is physically large, ethnically heterogeneous, has been colonized by the British and is a pluralist democracy. While the United States was just twice as well off as India in 1820, at that time though on a trivially small population base, in 1989 she was at least 15 times richer relative to India in 1989, and over 30 times richer in 1989 relative to India in 1820. By 1989, her population had also increased massively. An important factor has been the industrial revolutions that the United States has engendered and participated in. The United States is an important role model for India today.

Next, let us take Japan. Like India, she is an Asian country. She came to industrialization very late. As I will discuss in Chapter 5, Japan is the first case of late industrialization and catching-up in the twentieth century. She participated in her own industrial revolution. Therefore, Japan is an important role model for India today. Between 1950 and 1989, after suffering ravages in the Second World War, Japan has increased her relative per capita income index to a figure of 3,130 relative to India in 1820. Japan is now over 30 times better off than

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India was then. Relative to herself in 1950, she has increased her per capita income ten times. Relative to India in 1989, Japan was about 15 times better off. At that time, she was also 25 times richer than she had been in 1820, using Indian income of 1820 as the base case. Yet, in 1890 she was scarcely better off than India in absolute terms. Today, Japan's per capita income matches that of the United States. The lesson for India from the Japanese example is straightforward. Late industrializing nations can rapidly catch up with the rest if the appropriate lessons from successful participants in the industrial revolutions are followed.

Leaders start suffering from atherosclerosis. Witness the falling behind of Germany and the United Kingdom, relative to Japan, by 1989. Late industrializing nations can become leaders by leapfrogging over extant leaders by adopting appropriate strategies. There are many learning options available to India based on the American, Japanese, South Korean, and Taiwanese experiences. The issue is can the Indian elephant play leapfrog? Entrepreneurship and commercial and manufacturing activities in India have long histories, stretching back to the dawn of modern civilization. To understand modern Indian industry, it is important to know the history. The next two chapters deal with India's industrial past. In the real sector, however, path dependencies matter greatly because attitudes and approaches persist, even if insidiously, while bricks and mortar perish with the ravages of time. The path dependencies of India's industrial past may generate the psychological impulses propelling her economic future.

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3 Aspects of Indian enterprise history

Commerce and industry in ancient India

This history of India's entrepreneurial activity goes back to the dawn of civilization. India has been a commercialized country for over 4,500 years. Unlike in the United States, in which there was no industrialization at all prior to the eighteenth century, India had been an industrialized country thousands of years ago as well. Then there was a hiatus and only in the last 150 years or so has India resumed the path of industrialization. The story of the evolution of Indian enterprise is fascinating, since the time-paths of this particular process reflect the ebbs and flows of generic human civilization in one land mass.

In the 1870s, when Major General Alexander Cunningham, the Archaeological Surveyor to the Government of India,[1] published details of the first Harappan seal, which had been discovered at an ancient site in Harappa in what is now the Pakistan part of Punjab, the world become aware of what might potentially be India's past. Not until the early 1920s, when archeologists Daya Ram Sahni and Madho Sarup Vats discovered the city of Harappa, and Rakhal Das Banerjee discovered the city of Mohenjo Daro in Sind, both on the banks of the Indus, were the secret of India's past civilization finally revealed. The then Director General of the Archaeological Survey of India, John Marshall, stated that: "India … must henceforth be recognized … as one of the most important areas where civilizing processes were initiated and developed."[2]

Since then, the secrets of India's past have been revealed in detail. The Indus Valley Civilization, as evidenced by the Mohenjo Daro, Harappan and related finds, such as at Lothal, have been classified as dating back, at least, to 2500 BC. In the Indus Valley the rich alluvium soil was ideal for agriculture. The land produced plenty of surplus food, which allowed many persons to withdraw from food production, leave the land, and become urban dwellers. They then occupied

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themselves with craft, trade, religion, and politics.[3] This development of commercial and political cultures led to the development of institutions. An aspect of such institutional development was the emergence of standardization. Standardization has been referred to as a process by which the United States became a great manufacturing power at the end of the nineteenth century. This process had been discovered over 4,000 years before that.

The enforcement of standards led to the stabilization and homogenization of the cultural and economic practices of the Indus Valley Civilization. The level of conformity required was rigid and comprehensive. From pottery vessels, stamp seals, the script, the dimensions of bricks, the system of weights and measures, and norms for urban planning, absolute uniformity was maintained throughout the Indus Valley Civilization area for many centuries.[4] Trade and industry were regulated by standardizing weights and measures, and the diffusion of systematic production techniques. Bricks were made everywhere to only one ratio; the width would be twice the height and the length twice the width. There were three standard sizes. The most common dimension was 30 centimeters by 15 centimeters by 7.5 centimeters.[5]

The Indus people cultivated crops such as wheat and barley and grew pulses, sesame, linseed, several fruits, and cotton. Several animals, such as cattle, buffaloes, goats, sheep, domestic fowl, pigs, donkeys, and elephants, were domesticated.[6] Production of agricultural as well as industrial items was for trade and not just domestic consumption. There were large craft industries where the output exceeded local needs and was meant for export. Commercial production was organized in the Indus Valley Civilization by concentrating different trades and manufacturing activities in different city zones and encouraging specialization,[7] highlighting the fact that spatial and agglomeration economies, a concept popular today, had been conceptualized and related policies implemented 4,500 years ago.

A bead factory has been excavated at Lothal, an Indus Valley Civilization trading station in Gujarat, and an industrial area for copper working has been found in Harappa.[8] Indus Valley Civilization craft products were traded in Sumeria and Mesopotamia. Mesopotamia and the Indus Valley Civilization were the literate and urbanized societies of ancient times, representing the zenith of the Bronze Age. Mesopotamia was also agriculturally rich, as befits a river valley civilization. It had large quantities of grain, oil, and woolen textiles to offer.[9] The trade

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contact between the Indus and Sumerian civilizations were of ancient origin and by the 2370 BC to 2100 BC period was well established. It continued to be active until about the second quarter of the second millennium BC.

The Indus Valley Civilization centers of business were involved in maritime trade with centers of business in Mesopotamia through the sea lanes of the Persian Gulf. The commodities imported by Sumeria and Mesopotamia included timber, copper, carnelian, gold dust, lapis lazuli, and birds such as peacocks.[10] Also, a group of Indus Valley Civilization traders had settled in Sumer, as Indus-style engraved seals and other artifacts have been found there. One seal has the imprint of coarse cloth on its back. Thus, cotton was a major item of export from India. Wild cotton had been domesticated by the Indus Valley Civilization, and Indian weavers' and dyers' skills were quickly in demand from the neighboring civilization.[11] The Indus Valley Civilization traders used land and sea routes for trade. They carried on extensive overland trade through Baluchistan, using caravans of pack oxen or donkeys. Clay models of ox carts are common finds at Indus Valley archeological sites. These prehistoric Indus carts were heavy and lumbering vehicles, with solid wheels and a wheelbase of a meter. They resemble the ox carts found in the Sind region in the twenty-first century![12]

For river and sea transport, the traders of the Indus Valley Civilization used sailing ships. If shipbuilding and naval construction is to be taken as a major industrial activity, then the ancient Indus Valley Civilization was, by the standards of the day, a very advanced industrial civilization. Large harbor facilities have been discovered at Lothal, in the state of Gujarat, near Ahmedabad, where a very large dockyard and a wharf have been excavated. The Lothal dockyard had been ingeniously designed with a spillway and a locking device. It had artificial channels connecting it to a tributary of the Sabarmati, a river which flows through Ahmedabad. Several stone anchors were found in the dock basin. The wharf, constructed of brick, stretched over the entire eastern side of the town. Alongside it ran a high platform on which were located warehouses and granaries. The dockyard resembles the modern port of Rotterdam, albeit on a smaller scale, but had been built several thousand years ago.[13]

Finally, evidence has also been uncovered of technology transfer in ancient times, with Mesopotamia supplying metal and metallurgical skills to the Indus Valley Civilization, while it imported ships and

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shipbuilding skills. Mesopotamia had large quantities of silver and lead, and the Indus Valley Civilization was a large importer of these items. It also needed the associated metalworking skills.[14] On the other hand, a lack of good wood meant that Mesopotamia could not build large seagoing craft. The only vessels would be reed boats or dugout canoes, which would be risky and ineffective in ocean voyages. The dry and arid lands of Mesopotamia did not support forests, and there was a lack of timber.

The Indus Valley Civilization, however, possessed timber supplies. It had become expert in making boats built of planks, as the evidence of the Lothal dockyard shows. The use of wooden planks made boats larger and deeper, permitted the use of larger sails, and opened up ocean commerce. Metal tools developed with the ores available in the Persian Gulf region and Mesopotamia, and the metallurgical skills available, could be adopted by entrepreneurs of the Indus Valley Civilization. They, in turn, could use these tools to do the advanced carpentry required to build large ocean-going craft.[15] The issue of mutual technology and trade flows between countries is a concept in vogue today. Over 4,000 years ago, a mutually reinforcing process of trade and technology transfers led to the diffusion of capabilities between civilizations. The increasing use of metal enabled more efficient modes of transportation to evolve, and this process benefited both civilizations.[16]

Industry and trade in Maurya times

By 1500 BC, however, the Indus Valley Civilization had declined. During the Vedic ages, which occurred between 1500 BC and 900 BC, there was a considerable development of Indian culture.[17] The social structure of India changed, with landlords emerging as major controllers of resources. Semi-hereditary merchant classes developed because this is when the caste system emerged in its now well-known form. Trade and industry further developed along large-scale and specialized lines. There was the elaboration of a complex financial structure. This period contained the seeds of several aspects of economic and social organization still found in India.[18]

Since the sixth century BC India has been noted as having a substantial and prosperous mercantile class who grew rich on foreign trade.[19] There was limited trading contact between the Greek and the Indian

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civilizations, until the establishment of the Achaemenid Empire in Persia in the sixth century BC. This stretched to Greece in the west and to India in the east, and the empire brought both Greeks and Indians in contact with each other.[20] The Indian product that the Greeks consumed most was cotton cloth. Cotton weaving had begun with the Indus Valley Civilization, and had reached a high degree of refinement. Cotton cloth was a major item of trade between India and Greece. Much of the trade was carried over land routes. Indian traders were active in Iran and Mesopotamia in the sixth and fifth centuries BC. It was apocryphally noted that a tavern in southern Iraq was run by an Indian businesswoman.[21] Sea trade between these countries, via the Red Sea and Alexandria, was equally important, initially being in the hands of Indian and Arab traders.[22]

Around the first millennium BC, several navigation texts also emerged to detail the international sea routes between the Red Sea and the west coast of India. These texts also dealt with coastal shipping, so that the east coast towns of India could be described for the benefit of seafarers. Astronomer and geographer Claudius Ptolemy gave the world a detailed knowledge of the Bengal coast.[23] Subsequently, Greek traders became a part of the bustle of Indian ports. Ancient Tamil poets would sing of the good ships of the Yavanas, as the Ionian Greeks were named, that arrived at their ports bringing cool and fragrant wines and returned laden with pepper.[24]

There were extensive international trade flows between India and the Roman Empire. There was even global migration and settlements in India. Excavations by archeologists at Arikamedu near Pondicherry have revealed Roman pottery such as Arretine ware and wine amphorae from the Mediterranean. Thus, in the southern part of India, by AD 1, in the time of Augustus Caesar, there was a colony of Roman traders and settlers who played an important role in the Indian and Red Sea trade.[25] As well as the Romans, Egyptian merchants came directly to India to obtain silks, cotton, spices, and jewels. There were Arab ships engaged in the coastal trade with India, and Indian vessels involved in the coastal trade of the subcontinent.[26] As Octavio Paz remarked: "India, in contrast, was always in communications with other peoples and cultures of the Old World: first with Mesopotamia, and later with the Persians, Greeks, Kuchans, Romans, Chinese, Afghans, Mongols. The thought, religions and art of India were adopted by many Asian peoples; in turn the Indians absorbed and transformed the ideas and

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creations of other cultures."[27] Similar trade exchanges enriched both India and the rest of the then known world.

On the domestic front, the Mauryan Empire, which ran its course between 321 BC and 185 BC, was a formidable economic force. By the time of the Mauryan regime, agriculture had displaced a subsistence pastoral economy. Extensive tracts of virgin lands were cleared for cultivation by the state and private farmers. Trade and industry had emerged as a major field of economic activity in Mauryan times. With the spread of urbanization, and the growth in the demand for goods and services, craftsmen were ubiquitous in towns. They gradually turned into small manufacturers and gained in wealth.[28]

Mining also received special attention in Mauryan times, because of its revenue and military potential. The state ran all mines, salt pans, and pearl and conch fisheries as state monopolies. Where the mines or fisheries were difficult to run, or of uncertain profitability, they were leased to private operators. This started the concept of private and public partnerships.[29] The Mauryan state was an important economic actor. State-owned enterprises were key entities and the state itself entered the market as a dominant player, selling the output of its numerous manufacturing units and mines, the yield of its forests, the harvests of crown lands, and the produce received through tax collections. These goods were stocked in state warehouses maintained throughout the empire, and were sold directly through royal officers or through private agents.[30]

During this period, trade increased and urbanization proceeded rapidly and prosperity expanded the Indian market. The creation of road infrastructure, various amenities along the roads, and improved travel security facilitated trade. There was also a lifestyle-based economy, with a high demand for luxury goods. The gradual monetization of the economy, the standardization of weights and measures, and appropriate trade regulations substantially invigorated the market.[31] Several centuries before the arrival of Christ, the existence of domestic as well as long-distance seaborne and overland trade had given India a distinguished mercantile character.

India as a medieval economic power

Again, there is a hiatus. The story of evolution of Indian enterprise and business comes to a stop. Or, rather, there is a chasm as we do

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not know much about the interregnum. Let us move several centuries ahead to the medieval period to assess India's economic vibrancy. The nature of India's economic situation cannot be assessed without understanding its overall structure of activity, and its political structure, since the modes of governance will have directly affected the institutional conditions and the incentives agents will have faced.

The Indian economy was divided into three elements. There was a world of great cities, which formed the base for external trade, traders and financiers who moved wealth around the country in the form of bullion and credit notes. Next, there was an intermediate economy of mass artisan production in small country market towns. There were also the petty traders and revenue farmers or revenue collectors who existed in every political dispensation. Thereafter, the main bulk of India was an internal agrarian economy.[32] This internal agrarian economy responded to changes in other elements, since it was affected by money supply and demand conditions. Yet it was the core of India. It always retained a great deal of autonomy. The village markets and local cattle fairs were spasmodically and poorly linked to the sub-continental economy.[33] The village economies, however, relied on the functions performed by small traders and peddlers in the distribution of commercial goods. Thus, the structure of trade was characterized by extreme fragmentation, multiple markets, and price instability.[34]

To astute nineteenth-century observers, such as legal scholar Henry Sumner Maine, India was a land of Teutonic village republics and a set of medieval feudal kingdoms.[35] According to Maine,

> the village communities are little republics, having nearly everything that they can want within themselves, and almost independent of any foreign relations. They seem to last where nothing lasts. Dynasty after dynasty tumbles down; revolution succeeds to revolution; Hindoo, Patan, Mogul, Mahratta, Sikh, English, are all masters in their turn; but the village community remains the same … If plunder and devastation be directed against themselves, they flee to friendly villages at a distance; but when the storm has passed over, they return and resume their occupations … This union of village communities, each one forming a separate little state in itself, has, I conceive, contributed more than any other cause to the preservation of the people of India through all the revolutions and changes which they have suffered, and is in a high degree conducive to their happiness, and to the enjoyment of a great portion of freedom and independence.[36]

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Analysis of India's southern cone, for the Chola period, between the ninth and thirteenth centuries shows that there was not much sign of dense economic activities based on structured markets and complementary production units.[37] Between the fourteenth and sixteenth centuries, the foundations of a mercantile economy were laid in southern India[38] as extensive population movement into the dry interior areas took place.[39] In the first half of the fifteenth century, very extensive commercial contacts between western Deccan ports and the Persian Gulf started.[40] By May 1498, when Vasco da Gama navigated the Cape of Good Hope and arrived in India, the port of Calicut, now called Kozhikode, was the center of a thriving international trade, and the communities involved included Europeans, who came overland via the countries of the Ottoman Empire, and Muslim and Jewish merchants from North Africa, Turkey, Persia, and Egypt. Vasco da Gama confirmed what the European historians had vaguely suspected or might have known, that India was capable of industrial manufacturing at a quality level surpassing the then-developed European economies.[41] India, and possibly other Asian countries, possessed important production skills and capabilities.

From Aden and Hormuz, Arab merchants carried luxury Ottoman and Mamluk goods such as copper, mercury, vermilion, coral, saffron, carpets, porcelain, tin, colored velvets, silver, raisins, opium, horses, and textiles to Calicut. Items purchased from India and locations further east were textiles, pepper, ginger, cinnamon, cardamom, tamarind, amber, clove, nutmeg, mace, sandalwood, cottons, and coconut. Italian merchants would then collect Ottoman and Indian Ocean goods from Alexandria, Beirut, Cairo, and Antioch and distribute them throughout Europe.[42] Traders and merchants based in western India engaged in large-scale wholesale dealings, with operations extending from Mocha in Yemen to China. These were stolid Hindu merchants who had the financial capability to singly buy up the entire cargo of a European ship with half an hour's bargaining and then make prompt cash payments for the stock.[43] Alexander Hamilton, a seventeenth-century sea captain, wrote about his experiences and mentioned Abdul Gafoor, a Muslim trader of Surat, who could supply 20 ships, each between 300 and 800 tons, with £10,000- to £25,000-worth of stock on his own account. After that foreign stock was sent away, he could further invest in similar quantity of stock, again on his own account, for trade in the domestic market.[44]

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The model of trade and commerce was different in northern India. There were a number of small merchants and traders in the bazaar who kept the agricultural economy functioning. They also controlled trade between close localities. These groups played a role in the functioning of the revenue system as revenue farmers.[45] Then there were the great merchants with a vital political role in state finances. These were important merchants, with cultural links to the smaller merchants. They had a role as Mahajans, or great traders, who traversed several regions. The Nagar Seths were the commercial elite,[46] being given the title of "Banker to the City," for their commercial services rendered originally to the Mughal emperor Jahangir.[47]

By the middle of the seventeenth century, a very highly developed network of bills of exchange, called *hundis*, was being serviced by firms in Surat and Agra to remit large sums of money across the country. These firms attained a strong coherence across urban centers and warrior fiefdoms of the Gangetic plain, in the Maratha domains of western India and also in the emergent powers of the south such as Mysore and Hyderabad.[48] By the early eighteenth century, several large banking houses had taken over the remittance function for taxation during the Mughal regime. These large merchant corporations had highly developed bookkeeping, credit management, and treasury capabilities. Their mechanisms to collect revenues and make transfers enabled coordination of extensive trade patterns between regions. From intermediaries they also became financiers of rural production.[49] They were then able to develop political capital as financiers of regimes. The firm of Jagat Seths, in Bengal, originally set up by Marwari merchants Manikchand and Fatehchand, belonging to the Oswal caste, was one of several firms set up by people from Gujarat or Multan, or people of the Bania Aggarwal community, many of whom were from Rajasthan.

From the seventeenth century onwards, a complex and integrated mercantile system developed.[50] This supported a great Ganges–Jamuna trade route that consisted of trade in cloth and grains, and had extensive transport and banking facilities.[51] Concomitantly, several older cities declined. Cities such as Surat, Ahmedabad, Masulipatnam, and Dhaka decayed. Cities such as Delhi, Lahore, Agra, and Burhanpur shrank in size. Several important new centers established themselves. These were Lucknow, Hyderabad, Benares, Mysore, Srirangapatnam,

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and Bangalore, which grew rapidly. Trading cities such as Mirzapur, Kanpur, and Baroda arose to service the new trades.[52]

As the Indian economy evolved, the second-level traders and merchants tried to gather dependent agriculturists and weavers within an organization approximating the condition of laborers in textile factories. Low labor quality and low levels of technology in weaving appeared, however, to have led to a contractual system based on piecework. Nevertheless, attempts to mass produce goods would have been associated with the growth of new centers of production directed at mass regional markets. Simultaneously, rich landowners also encouraged the settlement of agricultural laborers on their lands in order to raise the revenue potential. Traders and bankers also invested in the village enterprises of Rajasthan and western India. By the first half of the eighteenth century, rural markets, called *ganjs* in north India, an example being Forbesganj, and *pettas*, an example being Royapettah, in south India, were being created. Such markets also appeared in peninsular India. These factors contributed to the long-term redevelopment of commerce in India.[53]

In the eighteenth century, Asia in general, and particularly India, remained the world's major center of artisan production. European countries were important to Asians economies because they provided the silver imports which helped Asia's great kingdoms to expand and develop. Yet, in the eighteenth century, the industrial revolution in Britain shifted the balance of power between India and Europe. The Indian and Asian economies went from being producers and exporters of artisan products to exporters of agricultural raw materials. But, prior to that, India had been a medieval economic power.

Industry and political economy before colonization

During medieval times, India had been a dominant global textile exporter. Other industries were, however, limited. Luxury metal and mineral items and textiles had been produced for sale in many towns. Benares had a thriving set of workshops that made brass and copperware for religious or domestic purposes. These circulated widely. Not much more by way of industry abounded, at least in north India.[54] The Indian artisans, however, could not fully transition to factory production. This inability to attain operational scale led to their marginalization.

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But not all of India was bereft of manufacturing or industrial capabilities. One area which India retained competencies in was shipbuilding, a historical legacy spilling over from the time of the Indus Valley Civilization. The shipbuilders of Surat, around the time that the British arrived as traders in India in the seventeenth century, had a production repertoire matching European processes. The Gujarati shipbuilders could replicate British vessels in all respects, if required, and the shipyards were capable of building vessels in the range of 500 to 1,000 tons. These vessels were world class for the times. They were capable of sailing to China and Europe.[55]

In the south, by the late eighteenth century, the tables had turned. In 1782, the ruler of Mysore, Haidar Ali had died and in 1783 his son Tipu Sultan was enthroned. There was a battle for political power between the British and the French in India, and Tipu Sultan responded to French overtures by insisting on direct dealings with Versailles.[56] In 1785, an embassy had left Mysore for Constantinople. The same mission was ordered to proceed to Paris. But, delayed in Iraq, it was superseded by a separate embassy sent direct to France in 1787. This emissary vessel docked at Toulon in June 1788, and Tipu's 45-person mission proceeded to Paris overland. At that time, himself in a certain amount of trouble, Louis XVI was in no position to gratify visitors' political or military wishes.

Tipu's other requests related to capability acquisition so as to modernize the economy. He asked for seeds of flowers and plants of various kinds, and for technicians, workers, and doctors. When the mission left for home at the end of the year it was accompanied by munitions experts, gunsmiths, workers in the porcelain trade, makers of glass, watchmakers, makers of tapestry, linen weavers, two printers of oriental languages, one physician, one surgeon, two engineers, and two gardeners.[57] Tipu had identified the factor which lay behind European efficiency. Trade was important. To this end, Tipu established a state trading company, encouraged investors to invest in it and organized a network of overseas factories located around the Arabian Sea and the Persian Gulf. These were modeled on the European trading companies, and included commercial and military staff.[58] Command of the Malabar ports gave Mysore a ready outlet to the sea, plus control of the outward trade in pepper and timbers, and of inward trade in horses from the Persian Gulf.

Technology was important. Tipu sought new crops by experimenting with seeds and plants from all over Asia as well as from France.

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Around his palace at Srirangapatnam, the ground was laid out for botanical experiments. He was personally responsible for introducing sericulture into Mysore. Other factories in Mysore produced sugar, paper, gunpowder, knives, and scissors. Ammunition factories at Bednur produced 20,000 muskets a year.[59] The Mysore factories also made rockets, which the British had not encountered before. The British Ordnance Office was keen on this product, and technology of rocket manufacture was transferred to the Woolwich Arsenal in 1806.[60] Clearly, there was a part of India, albeit small, that was as close to the evolving frontier of the time as might have then been possible.

The history of India in the eighteenth century is one, however, of political crises. The cumulative indigenous changes of the Indian heartland reflected commercialization of the economy, coupled with the decline of the great Mughals, especially after the death of Aurangazeb in 1707. As both Hindus and Muslims prospered in Mughal service, and traders flourished, they began to separate themselves off as stable landlords through northern India. Prosperous yeomen farmers left the Mughal fold, assisted by the several moneylenders and merchants who could provide the economic basis for doing so. Commercial growth eroded the political power of the Mughals.[61]

During the waning of the Mughal hegemony, many politically autonomous groups emerged. Military entrepreneurs farmed revenues, engaged in local trade, and built up land holdings. They were closely linked to the merchant houses noted earlier. These merchant houses and firms were vital politically, as they could mobilize large liquid capital reserves since they participated in the national trade and credit chains. All groups of persons represented indigenous capitalism forms, and derived wealth from commodity trade.[62] With the decline of the Mughals, there was further development of powerful commercial forces in the regions, and these led to the emergence of additional entrepreneurship in power and money.[63] The Marathas had emerged as a political power during the seventeenth century. They consumed large quantities of grain, cattle, and cloth from the Gangetic plain acquired by trade. Mysore state consumed large volumes of cotton, northern Deccan wool and hides, Benares sugar, and Carnatic region cloth.[64]

Then there was substantial expansion of European production and trade in the eighteenth century, and the development of aggressive capitalism. This assertiveness was expressed in the approach and

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policies of the European companies in India from the 1730s, and by the East India Company in 1757.[65] There was relatively little domestic resistance to the creation of commercial and then political empires.[66] The Indian entrepreneurs often emerged as a comprador commercial class. The East India Company, hence, was provided with some of the capital, knowledge, and support for its activities by this Indian commercial class. Indian capital and expertise were drawn into a partnership with alien invaders, and Indian commercial groups became uneasy or maybe even willing collaborators in creating colonial India.[67] Specifically, with respect to the opium trade, which emerged as big business after 1770, the relationships between Indian traders and British private traders produced substantial profits. The Malwa opium operations involved Gujarati and Marwari traders, operating in the hinterland, financing cultivation, and collecting the produce, and Parsi merchants in Bombay shipping the opium to China with British collaboration.[68]

Transition to British rule and de-industrialization

The economic history of Indian enterprise is inextricably tied up with that of the British after 1757. The 101 years between 1757 and 1858, and then the 89 years between 1858 and 1947 represent a dark age for India. India was a British colony, but not a British settlement. I use Sumit Sarkar's classification of the economic context.[69] These are: of a mercantilist phase between 1757 and 1813, marked by plunder and monopoly trade; a phase, between the years 1813 and 1858, when India was opened to the aftermath of the first industrial revolution and the impact of free trade; and the phase thereafter, from 1858 till 1947. In the era of direct British governance, financial and industrial capitalism took hold, and Indian industrialists and Indian financiers emerged.

The British were in India to exploit her and not develop her. Benita Parry remarks that "Colonialism was, first, last, and always an instrument of capitalism."[70] The narrative of the strategic mission changed over time from one of maximizing the profit margin to one of bringing civilization to a heathen land. An almost metaphysical obligation to rule subordinate, inferior, or less advanced peoples developed among colonialists such as the British.[71] Such a narrative was expressed in florid rhetoric, replete with boasts about the British as a governing

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race, with claims to a secular and divine destiny ordaining them to bring technological progress, reform of social practices, and rational government, in other words modernity, to a benighted people.[72] In this respect, journalist Mark Tully's words are important. He wrote:

> At least four generations of my mother's family before me spent their working lives in India. Most of them went "home" to retire, and I can well understand why. In the first place, they were positively discouraged from "staying on." Britain ruled India but did not colonize it. The British Raj did not encourage settlers. Settlers might set a bad example and "go native," hence destroying the carefully nurtured image of the difference, and hence the superiority, of the British race. The Raj, it was thought, depended on that image of superiority to enable it to rule so many with so few.[73]

Development in India occurred accidentally, or when the political economy situation tilted the balances in Indians' favor. Such a point was expressed by administrators, such as Shankar Nath Maitra, an ICS officer, who wrote:

> Whatever advance was made in the political or administrative field, there was hardly any change in the basic colonial economic pattern of the Raj till its last day. In fact it may be argued that the stagnation in economic field led to the excessive importance being given to politics and the capture of power. The administration especially in the Punjab was all for the *kisan* and all against the native *bania* but could say little against the British *bania*.[74]

The British went to India as traders in the early seventeenth century. The late historian Samuel Martin Burke, ICS,[75] noted that by 1740 British imports from India accounted for 10 percent of the British gross domestic product.[76] By 1757, after the battle of Plassey, they had consolidated their political position so much that they became the rulers of India. With the decline of the Mughals, from 1707, the consumption capacity of the Mughal elites diminished. Nevertheless, the consumption ability of the new political entities made up for the demand decline, as did foreign demand. The Marathas, the state of Mysore, petty chieftains, revenue farmers, and yeoman farmers supplanted the Mughal court as consumers. Repression was replaced by regression. The Aurangazeb regime had been brutally fascist. There was considerable political anarchy in India after Aurangazeb's death. India was back where it had been in the twelfth century, a bundle of

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feudal territories with neighbors at odds with each other.[77] This permitted the British to politically dominate India from 1757.

The British had been sucked into the Indian maelstrom by the dynamic of its political economy as well as their relentless drive for profit. In this process, the collaborating native business elites played a crucial role. Via these intermediaries, British businessmen joined the entrepreneurial class of late Mughal India.[78] The British then fused military and commercial power to achieve, on a larger scale, what the Indian local rulers had aimed for in the seventeenth century, the commercialization of political power. This process was carried out by a slow and steady penetration of the lines of power and flows of traded commodities, with settlement primarily based on silver. The bullion could be used to acquire or market political power. There were threads of continuity between the political economy strategy of the rulers of colonial India before she was colonized and the political economy strategy of the East India Company.[79]

The profit margin was the main British interest in India. The East India Company was out to make money.[80] In the expansion of the western economies, profit, and hope of further profit from trade, had been important, as the attractions of spices, sugar, slaves, rubber, cotton, opium, tin, gold, and silver over centuries testified.[81] Politics and governance were a means to an end. The greed motive was uppermost at the battle of Plassey, when Clive and his acolytes visualized that Mir Jafar,[82] whom they were installing as the Nawab of Bengal in place of Siraj-ud-Daula, could lay his hands on £40 million as soon as the British placed him on the chair, and salivated that the British would get a large slice of that sum. As it turns out, this was an erroneous assumption. Nevertheless, the presents received by the British were extraordinarily generous.

Clive headed the list of beneficiaries with a present of £211,500.[83] In addition, Mir Jafar had to pay £275,000 to the army for its troubles. Clive took another cut as a revenue grant of £27,000 per annum paid to him for the rest of his life, until his suicide. When, after 13 years, a House of Commons committee investigated the goings-on in Bengal, it concluded that presents to Britons in 1757 amounted to £1,238,575, and between 1757 and 1765 sums exceeding £2 million were disbursed to individuals in bullion and jewels.[84] There is considerable justification for Tapan Raychaudhuri's statement that: "The British school child who reads in his history text book of Lord Clive's

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wonderful achievements should also be told that the man was a criminal by any definition."[85]

Behind the British involvement in Bengal, via Plassey, had been the growth of Bengal's industry. The rise in textiles exports from Bengal between the late seventeenth and mid eighteenth centuries, and profits to be made in the trade between India, China, and Britain, had made Calcutta important since its establishment in 1690. Bengal was the wealthiest province in Mughal India. It was surplus in rice. Its weaving industry produced a high level of output till the late eighteenth century. It had a large number of literate gentry adept in revenue management.[86] The East India Company had been aware of Bengal's productive potential, and its contribution to international trade. It was India's granary; it supplied India and the world with muslin and silk yarn; and it had abundant saltpeter. These three items made it very important for trading companies.[87] Adam Smith had noted that the economic capability of Bengal derived from an extensive home market based upon inland navigation.[88]

These facets came to the notice of the British. They developed good relationships with the banking houses of Bengal, especially the Jagat Seths, and another banker Omichand.[89] The Jagat Seths and Omichand were useful intermediaries between Clive and Mir Jafar, ensuring that Mir Jafar would not respond to British attacks at Plassey and the British would be able to win the battle against Siraj-ud-Daula. Clive's victory in 1757, installing Mir Jafar as the ruler, delivered the revenues of Bengal into the hands of the East India Company. Before Plassey, the East India Company had to pay for its goods with imported silver.[90] After Plassey, the new annual proceeds of the Bengal revenues paid for the goods acquired from Bengal and other parts of India, and bullion from external sources was not required.[91]

Earlier, British and other European traders had to regularly bring bullion into India, as Indian cotton and silk goods had a flourishing European market while Indian demand for European products like British woolens was slim. After Plassey, bullion was rarely imported by the trading companies or brought into Bengal from other parts of India in any quantity.[92] The Bengal revenues amounted to £4 million a year. Thereafter, the East India Company began to create its own state using these Bengal revenues. The "investment" that the British made in acquiring Indian goods for sale overseas was financed out of the territorial revenue amounts until 1813, when the East India Company

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lost its trading monopoly, and its territorial and commercial accounts were separated.[93]

The economic consequences on Indian enterprise of the battle of Plassey were severe. In 1783 the Select Committee of the House of Commons set forth the position in their Ninth Report:

> A certain portion of the revenues of Bengal has been, for many years, set apart for the purchase of goods for exportation to England, and this is called the "Investment." The greatness of the Investment has been the standard by which the merit of the Company's principal servants has been too generally estimated; and this main cause of the impoverishment of India has been generally taken as a measure of its wealth and prosperity. Numerous large fleets of large ships, loaded with the most valuable commodities of the East, annually arriving in England in a constant and increasing succession, imposed upon the public eye, and naturally give rise to the opinion of the happy condition and growing opulence of a country whose surplus productions occupied so vast a space in the commercial world. The export from India seemed to imply also a reciprocal supply, by which the trading capital employed in those productions was continually strengthened and enlarged. But the payment of a tribute, and not a beneficial commerce, to that country, wore this specious and delusive appearance.[94]

The takeover by the East India Company of revenue collection and administration, first in Bengal, and then successively in the south, the Gangetic plain, and western India, led to disruption in the pattern of external trade and changed the relationships of entrepreneurs with the new state.[95] A rent extraction for over 50 years, as in Bengal, is bound to have deleterious consequences. Simultaneously, the use of harsh fiscal means, such as taxation and personal rent extraction, and corruption of the Company's servants precipitated economic decline.[96] The stories of nabobs, who returned to Britain from a career in India and purchased enormous country houses and seats in the House of Commons, are legion.[97]

Indian merchants were excluded from free trade in many valuable commodities, with such exclusions enforced by military power. After the battle of Buxar in 1764, when Mir Qasim, the successor of Mir Jafar, was removed from power, British traders became monopolists and the largest purchasers of goods. The East India Company also acquired the sole rights to revenue collection, or the *diwani* rights, in 1765. The net surplus accruing to the East India Company was about

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50 percent of the revenues they collected on behalf of the Mughal emperor for the regions of Bengal, Bihar, and Orissa.[98] The change in the pattern of incomes and expenditures of the East India Company after 1757 had considerable implications for the internal economy as it started the drain of India's wealth. Nicholas Dirks writes that:

> Reinvestment, in other words, was nothing other than the use of the "tribute" to support the colonial state itself. But the colonial state not only did almost nothing to invest in infrastructure or provide meaningful administrative services, for agrarian or any other purposes; it was in fact designed entirely to extract resources from India for the enrichment of the Company in particular and Britain more generally. In the end, the argument over how to interpret the economic data concerning the impact of the British presence on India hinges on whether one views the colonial state as legitimate and benevolent, or as fundamentally extractive in a way no indigenous state could be.[99]

As early as 1772, Alexander Dow, in his *History of Hindostan*, had asserted that Bengal's decline had commenced with the battle of Plassey, and that it was a direct result of foreign dominion. He calculated that Bengal lost approximately £1.5 million each year through extraction of bullion and the use of monopolies in inland trade, especially in basic commodities such as salt, betel nut, and tobacco.[100] In 1888, John Strachey, who authored a work on the finances and public works of India, explained the mechanism of wealth drain: "The Secretary of State draws bills on the Government treasury in India, and it is mainly through these bills, which are paid in India out of the public revenues, that the merchant obtains money that he requires in India, and the Secretary of State the money that he requires in England."[101] The emasculation of Bengal's textile industry was graphically described, at the turn of the twentieth century, by journalist William Digby.[102] He estimated that the population of Dacca dropped from 200,000 persons to 79,000 persons between 1787 and 1817, while exports of Dacca muslins to England, which had amounted to ₹8 million in 1787, were zero by 1817.[103]

The home charges, payable to Britain for defense and administrative services by India,[104] and private remittances, were funneled through Indian exports. The drain of wealth was expressed by India's export surplus. This surplus became vital for Britain's balance of payments in the nineteenth century. The nationalist leader Dadabhai Naoroji[105] argued that the amount drained away a huge sum which would have

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materially improved Indian welfare if invested in the country.[106] The historian Sarvepalli Gopal remarked that: "The British had given an impetus to the destruction of the old economy but did not permit the rise in its place of one more suited to the modern age. India was made to serve as the supplier of raw materials to Britain's new industries and as the market for her manufactured goods."[107]

In the rest of India, too, commercial considerations drove British conquest. The East India Company's conquests in the western part of the Deccan peninsula and in central India arose primarily because of its commercial needs. Since the early 1780s the Bombay cotton trade had boomed. After 1784, the Company needed ever-increasing quantities of raw cotton to sell in China to increase the quantities of tea procured from the Chinese. Control over cotton territories, in the area now called Maharashtra and Madhya Pradesh but then called the Deccan or central India, was a motivator behind the Maratha Wars. In these wars, the merchants of western India, primarily from Gujarat, who were intermediaries in the raw cotton trade between the East India Company and the rural hinterland of growers, were also financiers of the East India Company.[108] Again, the Indian commercial class had been drawn into a symbiotic relationship with foreigners in furthering the conquest of India.

A final point about the establishment of British economic and political power after 1757 is of conceptual and historical interest. A concern of contemporary innovation scholars is how the nature of financing affects innovative activities. Other than market demand, and the type and quality of institutions, does the quantity and type of financing available matter in generating innovations?[109] Therefore, did the funds available from Bengal, and then subsequently from the rest of India, finance the British industrial revolution? This line of thought had been suggested a century ago by Brooks Adams[110] and William Digby, who had noticed the close correlations between dates such as the 1757 battle of Plassey and the succeeding political events, and the subsequent emergence of innovations during the first industrial revolution. John Kay's flying shuttle was invented in 1760, James Hargreaves' spinning jenny in 1764 and Richard Arkwright's water frame in 1768.

William Digby wrote:

England's industrial supremacy owes its origin to the vast hoards of Bengal and the Karnatik being made available for her use. Had this happened

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honourably and in the ordinary course of trade it would have been matter for satisfaction. Before Plassey was fought and won, and before the stream of treasure began to flow to England, the industries of our country were at a very low ebb. Lancashire spinning and weaving were on a par with the corresponding industry in India so far as machinery was concerned, but the skill which made Indian cottons a marvel of manufacture was wholly wanting in any of the Western nations. As with cotton so with iron; industry in Britain was at a very low ebb, alike in mining and in manufacture. The connection between the beginning of the drain of Indian wealth to England and the swift uprising of British industries was not casual: it was causal.[111]

Perhaps, the closeness of the various dates is an amazing coincidence. Nevertheless, it will be enlightening to investigate, in depth, the relationships between the financing of the first industrial revolution innovations by the cash transfers made from India to Britain after 1757. Certainly, India as a market for British products was extremely important. But did the Indians, or rather the loot extracted by the British from India, finance the British industrial revolution? An estimate of the size of the financial drain from Asia has put it at 1.7 percent of Britain's gross domestic product in 1770 and at 3.5 percent in 1800. There was equally large wealth extraction taking place from the sugar plantations of the West Indies. When the Asian and the West Indian financial drain amounts are added together, they come to 86 percent of the amount of Britain's gross capital formation from domestic savings.[112] In other words, almost half of the financial sums available to Britain for making domestic and foreign investments came from the colonies.

The relationship between the wealth drain from colonies and the financing of the industrial revolution deserves detailed analysis. Britain had gained control over India at a stage when she was developing technologies that could add value to resources. Her motives lay in the supply by the colony of biological and mineral resources, while she supplied the colony with manufactured value-added products. This strategy would require both the suppression of competition in areas of Indian competence, such as shipbuilding and textiles, and financial rent extraction.[113] The use of Indian funds to fund innovations in Britain would, therefore, be consistent with this strategy.

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The plantation industries

The modern industrialization of India began with plantation industries such as indigo, tea, and jute. Justice Mahadev Govind Ranade, an eminent jurist and political economist, had observed that the first step in the progress of India from a purely agricultural country to a manufacturing country was due to the growth of agro-based industries, producing items such as indigo, tea, coffee, oils, sugars, beer, and tobacco. This growth was simultaneous with the growth of the jute, cotton, silk, woolen, paper and flour-milling industries. The ratio of India's manufactured to raw materials exports had become 30:70 by the end of the nineteenth century.[114] The first attempt to exploit India's natural resources, clearly for enhancing British economic well-being, came at the end of the eighteenth century when the British established indigo plantations in Bengal. The vegetable blue dye had been obtained from small indigo growers of Gujarat, but their cottage industry was extinguished by better-organized plantations in eastern India. The East India Company brought in experienced planters from the West Indies and rapidly expanded the Indian indigo output into a major industry. The industry flourished until 1897, when German chemists, during the second industrial revolution in which Germany was an important participant, developed the aniline substitute dye and destroyed the natural indigo dye market.[115]

Another item that had influenced British seeking of wealth in Bengal was tea. Tea was introduced to Britain in the second half of the seventeenth century. By the early eighteenth century it had become an article of mass consumption, along with sugar, coffee, and chocolate, and was the most important item in British trade with Asia.[116] In the eighteenth century all tea was grown in China. British demand for tea, however, was insatiable, and by 1757 it was over 3 million pounds per year. The Chinese initially accepted only silver bullion in exchange for tea. Gradually, they became interested in other Indian products such as cotton piece goods and opium.[117] To fund the British tea habit, large funds were necessary, and they were available in Bengal. It was, however, necessary for the British to diversify their sources of tea supply. As we know from the industrial organization literature, single-supply sources create hold-up problems.

In 1820 the tea bush was found growing wild in Assam but the Indian tea industry did not start until 1841, when a Dr. Campbell of

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the Indian Medical Service brought China tea seeds from the Kumaon region of the United Provinces, now called the state of Uttarkhand, and planted them at 6,000 feet near Darjeeling. In 25 years, there were 39 gardens covering 10,000 acres. By 1874, the Darjeeling area was producing 4 million pounds of tea a year from 113 gardens.[118] The development of tea plantations in Assam, Darjeeling, and southern India was strongly encouraged, though plantations in Assam did not commence till 1855.[119] The renewal of the East India Company charter took away its trading functions, but from then on Europeans were granted the right to own land in India. With this Charter Act of 1833, European settlement in India made possible direct control by the British over the cultivation of crops such as tea, coffee, and indigo.[120]

The growth of the tea industry in India in the second half of the nineteenth century then became phenomenal. It took place mostly in Assam, not only because of soil suitability and climate but also because large tracts of virgin land were available in upper Assam for settlement.[121] The original terms of the lease were attractive. The leases were for 99 years, a quarter of the land area was rent free in perpetuity, and the other three-quarters of the land area was rent free for 15 years. Thereafter, the rent rose slowly to peppercorn levels for the full term. Renewal was promised at moderate rates. As a result, thousands of acres were taken up by prospective planters and companies.[122]

That tea was an important business of the Raj and for the Raj, an industry intended to enhance British well-being and wealth, was revealed by the nature of relationships the industry had with the government. Tea estates varied greatly in size and conditions, but the industry projected a solid face to the world via its Indian Tea Association.[123] Under the Government of India Act of 1935, which devolved political autonomy to the provinces, the small Assam provincial legislature had 108 members. Of these, seven members of the Indian Tea Association represented European planting interests in the legislature. The Indian Tea Association group had direct representation in the central legislature in Delhi, where its representative from 1937 to 1947 was Percival Griffiths, a former ICS officer and later an important economic historian.[124]

The tea lobby clearly had substantial local political clout. The Governor of Assam kept in touch with the tea planters, and on tour stayed with the superintendent of big tea companies such as the Jorehaut Company or the Bishnauth Tea Company, rather than with

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civil service district heads, or in the district circuit house.[125] The close and symbiotic relationship between the tea industry and the local administration was taken for granted.[126] Shankar Nath Maitra, an ICS officer, wrote: "I was DC Cachar for two years and saw a lot of Nobby Clark, Secretary of the Surma Valley Branch of the ITA. He was proud of his position in the ITA and proud of his decoration CIE. Nobby Clark once frankly told me, 'I am not a planter, I am a diplomat,' and on occasions behaved as if he was the Government of Assam; particularly during the war, the ITA was an *imperium in imperio*."[127]

The indigo plantations of India have vanished, but tea plantations are still a vibrant and major part of India's industrial landscape. A third plantation industry that arose at that time, and is still very much in existence today, in reduced form, was jute. As a young professional in Calcutta, my first forays into the world of Indian industry, in the late 1970s, were to examine the relics of jute mills that had been taken over by the government; some of these jute mills really did have assets on their registers which dated back to the late 1800s. Some of these machinery items were still in operation. In one of the sick jute mills I visited, the mill's office areas were positively dark and Dickensian. There were high tables and high stools for the office clerks. The asset registers were three feet in width and two feet in height. They each required two persons to carry them around. The size of the tables required to fully open one of these registers can be guessed. These registers dated back to the formation of the mills. While the content had been updated with new assets, those who operated the accounting system clearly had never heard of information technology. Quill pens and ink had just been given up in favor of fountain pens!

Jute is a Bengali word. The plant is mostly grown in Bengal. It was cultivated well before the British came to India. Its fibers provided the ropes and cordage that other parts of the world made from hemp. Handloom weavers also made a rough cloth from it, and it was extensively used for local needs. With general growth in world trade, in the early nineteenth century, the export of homemade gunny bags increased, the consignments being sent to Southeast Asia and North America. As a cottage industry, however, jute manufacturing was inefficient and the product quality was poor. The first industrial revolution had not come to India and Bengal had at that time no coal or other fuel; the establishment of factories was out of the question.[128]

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In 1822, flax spinners in Dundee in Scotland experimented with jute manufacturing, but the results were unsatisfactory. In 1833, the British shipped home further jute fiber to supply the flax industry in Dundee.[129] The Douglas Foundry had developed a spinning machine that could handle jute as well as flax. By 1838, considerable improvement in technical processes had taken place, and Dundee factories rapidly took to using jute in place of flax. By the middle of the nineteenth century Bengal exported large quantities of raw jute to Dundee.[130] The Crimean War interrupted the supply of flax and hemp from Russia to Britain, which led to jute permanently taking their place and transformed India's manufacture of jute from the handloom to the powerloom level. The development of the Bengal coalfields in the middle of the nineteenth century made it possible to establish jute mills in Bengal.[131]

In 1854, George Ackland entered into a partnership with Bysumber Sen and in 1855, with the aid of a Dundee jute overseer, they started the first Indian jute spinning mill, at Rishra, on the banks of the Hooghly. For a time they also carried on handweaving. In 1859, the Borneo Company introduced powerlooms for jute cloth.[132] From this stage, development was rapid. By 1873, over 1,200 powerlooms were in operation. From 1872 to 1875, 13 new mills were launched and the number of looms had trebled.[133] By 1882, there were 20 jute mills, all but two in Bengal, which employed 20,000 persons. By 1885, Calcutta was the largest exporter of jute to the United States. In 1908, 38 companies employed 184,000 Indians and 450 Scotsmen in Calcutta.[134]

One of my forays as a young professional working in a consultancy practice in Calcutta was to the Hastings Jute Mill in Rishra. I am not sure that this was the mill set up by George Ackland and Bysumber Sen but it was reputed to be one of the oldest jute mills in India. This mill was very active in the late 1970s and is still in operation today. It was one of the best jute mills I had ever come across, though I had seen only a few jute mills and could not be called an aficionado. It was clean and professionally managed. Its manufacturing facilities had been continuously updated. This gave it excellent productivity, unlike that of other jute mills that had sickened and "died."

At the Hastings Jute Mill, over 4,000 persons worked in shifts. While the clatter of machinery was immense, and clearly the concept of information and communications technologies was in someone's imagination many decades away, the Rishra scene was reminiscent of

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a painting by L. S. Lowry of a Manchester landscape. Of course, only later in life did I understand what a Lowry painting evoked, but in the late 1970s there was an industrial India that had continued in the same way for over a hundred years. The Hastings Jute Mill also had an immense country house dating to the late eighteenth century in its huge compound, which reputedly had been a rest and recreation getaway for Warren Hastings and his Indian ladies, and each mill covered an area of about 100 acres.

The First World War, and the demand for sandbags, which is now permanent since there is a war on all the time somewhere in the world, stimulated the manufacture of jute. By 1919, there were 76 mills. By 1940, there were 110 mills, all highly profitable. An analysis carried out in 1927 showed that of 51 jute mills quoted, no fewer than 32 had paid dividends of more than 100 percent in one or more years since 1918, while 10 mills had never paid less than 40 percent in any of those years.[135] In spite of the initial partnership between George Ackland and Bysumber Sen, the industry had been primarily in British and American hands. Indians became major participants in the sector in ever-increasing proportions in the twentieth century. In 1927, about 50 percent of the shares were in Indian hands.[136] This process intensified in the fourth decade of the twentieth century. In 1937, over 63 percent of shares in 45 jute companies surveyed were owned by Indians.[137]

The industrial revolution industries

Meanwhile, what of the industries, such as coal, iron and steel, that had been the mainstay of the first industrial revolution in Britain from the mid to late eighteenth century onwards? Let me commence with coal because it is a short story for the moment, though the principal company in contemporary India, Coal India, has just made the largest-ever initial public offering of shares in India. The coal mining sector in India dates back, at the earliest, to 1774 when John Sumner and Suetonius Heatly of the East India Company were granted permission to work on seams found in the Raniganj area of Bengal. Several seams are supposed to have been worked on. Some coal extracted was delivered to the government in 1775, but this was of low quality. Then no further attempts to work any seams of coal were made till 1814. Again, this venture was unsuccessful. Yet, sufficient interest had been

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aroused for other ventures to be started between 1820 and 1825.[138] Nevertheless, the growth of Indian coal mining remained sluggish for want of demand. There was the odd steam-driven jute mill, which used powerlooms, but only the introduction of steam locomotives in 1853 gave rise to serious demand for coal.[139]

In the iron and steel sector, the tables have now been turned. A company run by an Indian, the Arcelor-Mittal group, is on its way to becoming one of the world's leading iron and steel producers in the twenty-first century. The manufacture of wrought iron in India has a long history. The famous iron pillar in Delhi has resisted rusting for over 2,000 years. India had world-class manufacturing capabilities in iron and steel several centuries before the Europeans and the Americans acquired theirs. Percival Griffiths[140] noted that the Damascene swords, which were famous during the Middle Ages, were originally forged in Hyderabad.[141] Steel made in the furnaces at Konasamundram and Dimdurti, two places situated in modern-day Andhra Pradesh, attracted buyers from other parts of Asia. Manufacture of the swords from the steel was carried on in widely scattered localities. These swords set the standard for iron work. They were made in Damascus, Isfahan, and in several places in southern India.[142] Incidentally, Hyderabad is still famous today for gun metalwork.

The manufacture of iron items, certainly by traditional village methods, was well established in India in the eighteenth century. It was largely a hereditary tribal craftsmen trade.[143] While the methods used were simple, and the iron usually impure, the nature of the work suggests that the craftsmen were more highly organized than their counterparts in other industries. Apparently, from the late eighteenth century onwards, European entrepreneurs had tried to improve local iron production by splicing in isolated pieces of British technology, such as the use of smelting coal and blast furnaces.[144] The first modern exploration of iron resources was carried out in the Madras presidency in the early nineteenth century, but no practical results emerged.

A more determined attempt was made by Josiah Marshall Heath, a Madras civil servant who had become interested in the local iron workings at Salem. This is first known instance of a civil servant turning to industrial entrepreneurship, though after independence many ICS members became industrialists,[145] an administration-to-industry career transition still in progress. Encouraged by Thomas Munro, then the Governor of Madras, J. M. Heath retired from the civil service and

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studied the iron industry in England for four years. In 1830, with the aid of English workmen and machinery, he established an iron works at Porto Novo, a location south of Pondicherry, on the Madras coast. He became a textile manufacturer and contractor for the East India Company, so as to initially finance the iron works from the profits of his cloth dealings. While J. M. Heath was a visionary, he was not a good businessman. He lost money in both textiles and iron. Several times, the government rescued him financially. From 1850 there was less help, and by the early 1850s the works closed.[146] The company was finally wound up in the 1870s.[147]

The first modern iron works in India was the Bengal Iron Works Company of 1874 at Barakar. The company started producing iron in 1877, but closed down two years later. The Government of Bengal bought the defunct firm, operated it for a few years and then sold it to a group of British businessmen in 1889. It was renamed the Bengal Iron and Steel Company. In 1897, the government agreed to purchase 10,000 tons of iron annually for ten years at rates 5 percent below the import price. In 1910, the company got access to new supplies of iron ore and coal, and became a producer of iron products. It, however, lacked expansion potential and ceased effective production in 1925.[148] In 1918, a second iron works had been founded in Bengal by the Indian Iron and Steel Company. This was linked to the Bengal Iron Works Company via managing agents Martin Burn and Company. The Indian Iron and Steel Company began production of pig iron in the early 1920s. The two companies were amalgamated in 1936. A diversification move was made by setting up the Steel Corporation of Bengal in 1937. This plant produced steel from iron from the Indian Iron and Steel Company, and by 1945 supplied about 20 percent of the market.[149] Eventually, in the 1970s, the Indian Iron and Steel Company became sick and later defunct.

Was there no hope for heavy industrialization by Indian entrepreneurs? In an often quoted passage, an American economist, Dennis Buchanan stated:

> Here was a country with all the crude elements upon which manufacturing depends, yet during more than a century, it has imported factory-made goods in large quantities and has developed only a few of the simple industries for which machinery and organization had been highly perfected in other countries. With abundant supplies of raw cotton, raw jute, easily mined coal, easily mined and exceptionally high grade iron ore; with a redundant

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population often starving because of lack of profitable employment, with a hoard of gold and silver second perhaps to that of no other country in the world, and with access through the British Government to a money market which was lending large quantities of capital to the entire world; with an opening under their own flag for British business leaders who were developing, both at home and in numerous new countries, all sorts of capitalistic industries; with an excellent new market within her own border and near at hand in which others were selling great quantities of manufactures, with all these advantages India, after a century, was supporting only about two percent of her population by factory industry.[150]

The next chapter sets the record straight with a recounting of several Indian accomplishments in the emergence of modern Indian industry.

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4 *The emergence of modern industry*

Entrepreneurship in the nineteenth century

In 1976, in an obscure corner of the Rajabagan dockyard of the Central Inland Water Transport Corporation, in the southern reaches of the Hugli, the river that flows through Calcutta, a huge pile of rusting junk metal was discovered. On closer inspection, it turned out to be a steam engine. It was one of the first steam engines to come to India, and it might even have been the first. It had belonged to the India General Steam Navigation Company. Prior to that, it had belonged to the Calcutta Steam Tug Association.[1] The identity of that scrap metal's original corporate owners is not important. The identity of the companies' ultimate owner is. The owner of the companies was Carr, Tagore and Company, a firm founded by Dwarkanath Tagore.

Dwarkanath Tagore, known as the Prince, has gone down in history as the grandfather of two extremely illustrious grandchildren. The older of the two, Satyendranath Tagore, was the first Indian to join the ICS in 1863. A younger grandchild, Rabindranath Tagore, was India's first Nobel laureate, winning the literature prize in 1913. Dwarkanath Tagore, however, deserves approbation as the father of modern entrepreneurship in India, in the form that we know it. His firm, dating back to the 1820s, Carr, Tagore and Company, was what we today classify as a holding company for a conglomerate of businesses. It had interests in many activities, and particularly in shipping. Dwarkanath Tagore voyaged to Britain in his own ship. Carr, Tagore and Company had also formed the Bengal Tea Association, which was the first Indian enterprise to start tea cultivation.[2]

In 1836, a company operating tugs went out of business in Calcutta. Dwarkanath Tagore purchased their steamer and set up the Calcutta Steam Tug Association. Just a month earlier he had purchased the Raniganj Colliery. This colliery had been set up by one William Jones, not to be confused with the orientalist of the same name. This William

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Jones was an engineer who had arrived in Calcutta in 1800 and, after working for others for a decade, had set up a canvas manufacturing factory at Howrah. He then set up a coal mine in Raniganj, and was the first person to use steam power to drive the engines that drained the mines.[3] The mine was, however, unsuccessful and was bought by Dwarkanath Tagore. Dwarkanath Tagore subsequently bought up a number of other collieries, and his steamer business was then assured of a reliable supply of coal. This is the first known instance of an Indian business engaging in a vertical integration strategy. The steamer and the coal business, which also supplied the steamers owned by the government and a few local factories, were the basis of a large commercial empire which survived into the mid nineteenth century.

Much earlier, in 1779, a Colonel Watson had set up two large windmills, each 114 feet high, in the Watgunge area of Calcutta. Industry in India that was not based on animate energy, generated from animal or human power, is assumed to have commenced from that time. William Carey, Joshua Marshman, and William Ward have gone down in history as the first missionaries in India. They, however, also deserve a place in the industrial pantheon. At Serampore, near Calcutta, where they set up a college, there was a small industrial complex that included a type foundry, printing presses, and a paper factory. These were run on steam power, and the complex is believed to have been the first to have used steam power for any manufacturing activity in India. In the engineering activities at Serampore, guidance had been given by the engineer William Jones.[4]

Meanwhile, in the 1820s, the Bowreah Cotton Mills had been set up by British interests at Fort Gloster, a location 15 miles south of Calcutta in the Howrah district. The mill used steam power, and had brought over several British women to impart machine-spinning skills to local staff. Some of these women, apparently, moved on to easier, if less virtuous, means of livelihood![5] The Bowreah Cotton Mills closed down in 1837, and then reopened as the New Fort Gloster Mills Company. In this company, Dwarkanath Tagore was a major shareholder. By 1840, the mill was the heart of a large industrial complex that included a cotton twist yarn factory, a rum distillery, a foundry, an oil mill, and a paper mill. There were five steam engines in the complex.

Dwarkanath Tagore's and others' subsequent efforts, in the first part of the nineteenth century, led to the Indians' adoption of some of the

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products of the industrial revolution, and also the emergence of modern large-scale factory enterprise.[6] Institutionally, important changes had also occurred. The first Companies Act in India was passed in 1850. A second Companies Act of 1857 allowed limited liability for companies in all areas, including manufacturing, banking, and insurance.[7] Wherever large capital investment was necessary, the advantages of a limited liability joint stock company were obvious. The limited liability law introduced the era of modern business enterprises into India.[8] By 1860, several companies were registered.[9]

Meanwhile, in western India remarkable changes were under way. In the cotton business, Indians had long established a substantial business foothold. The Crimean War had been, as noted, good for the jute business. It was also good for the cotton business. What was even better for the cotton business was the American Civil War between 1861 and 1865. At one fell swoop, the Lancashire mills lost their supplies of long staple cotton from the southern part of the United States. This vacuum was immediately exploited by Indian merchants, and later British merchants based in India, who made substantial fortunes.

Transfer of capabilities between businesses is an important topic today. Yet, that is precisely what happened in nineteenth-century India. The Indian cotton trading fraternity used their finance-raising capabilities, wealth, connections, trade and market contacts, knowledge of markets, and familiarity with the business to venture into the manufacture of textiles. The experience that many prominent Indian merchants had acquired, of a range of geographies and commercial approaches, facilitated the transfer of their abilities and knowledge from foreign trade to domestic industry.[10] In Bombay and Ahmedabad, this accumulated financial, social, and knowledge capital was used to create a genuinely indigenous textile industry. Raw cotton was readily available from the hinterland that had been opened up by the railways constructed from 1853 onwards. The British showed little interest in investing in an industry which would compete with Lancashire.[11] Conversely, British trading houses found that by faster adoption of many feasible business functionalities, because of the arrival of the railways, they could beat the Indian cotton traders at their own game.[12]

The modern Indian cotton textile industry was born in 1854,[13] when a cotton mill powered by steam was established by Cowasjee N. Davar in Bombay.[14] There were 10 cotton mills by 1865. By 1875, 47 cotton mills were in operation, and 79 mills by 1883. The expansion of the

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Bombay industry succeeded in replacing British yarn exports to China in the 1870s and 1880s. The products of Lancashire had dominated Indian yarn and cloth markets until the 1870s. The revival of Indian textile production led to a revival of Indian fortunes.[15] In 1880, there were 58 cotton mills, employing 40,000 persons. Most of the mills, and 80 percent of the workers, were in Bombay and Ahmedabad.[16] When difficulties arose in the China market in the 1890s, Indian mill owners responded by creating integrated spinning and weaving mills. The number of looms in Bombay doubled between 1900 and 1913.[17] The diversification into textile manufacture was by traders. In Bombay, the cotton dealers had sought a new business to diversify their capital. They found it in textile manufacture. This could smooth profits when commodity markets became volatile.[18] Yet, new investments in automatic looms were constrained by problems of labor discipline, since workers held back on effort so that investments were not lucrative for owners.[19]

Most of the pioneer cotton mills are gone. In Bombay, these belonged to groups such as Petit, Wadia, Tata, Currimbhoy, Sassoon, Khatau, Goculdas, and Thakersey. Among the British business houses were Greaves Cotton and Company, W. H. Brady and Company, and Killick Nixon. In Ahmedabad's textile industry, the prominent names were Sarabhai and Lalbhai. Today, Sanjay Lalbhai, a former classmate of mine, still runs Arvind Mills, one of India's remaining operational integrated textile mills. In Ahmedabad, the old established trading and banking groups could finance a strategy of outsourcing whereby artisans would be provided with spun yarn to weave cloth that could then be marketed by the trading groups. As a consequence of the close integration of trading, financing, and manufacture within the city, and inside family groups, the Ahmedabad textile industry grew fast between 1900 and 1913, with a distinctive profile as the supplier of high-quality cloth.[20]

Grassroots industrial entrepreneurship, or what is described as the democratization of entrepreneurship, could not occur in India in the nineteenth century. Industrial development capital was severely short, and persistently scarce in India. Investments in industry cannot happen without funds and credit. In the nineteenth century, the banks in India, implicitly supported by government, were conservative and extremely choosy with lending.[21] Given this situation, it is not surprising that the Indian pioneers in large industries came from communities

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that had accumulated capital in trading and banking activities. They could raise money relatively easily.

Yet, a lacuna remained. Having transferred commercial capabilities to the sector, the Indian industrialists stopped there. Technology transfer was forgotten. The development of India's textile industry was characterized as a process of relentless improvisation in the use of old machinery, the manipulation of raw materials, and the exploitation of cheap labor.[22] The Bombay industrialists were relatively slow to adopt new production technologies in the nineteenth century and stuck to an inappropriate and less productive type of spinning machinery for much longer than Japanese rivals.[23] This led the Indian textile industry to be extremely unproductive. Conversely, now that Lancashire had been bested by Bombay and Ahmedabad in textile manufacture, the British became more successful than the Indians in cotton trading. After 1870, an increasing proportion of the profits in the cotton trade went to foreign firms, while previously the greater part had been earned by Indian traders.[24] By 1875 the relative roles and importance of the British and the Indians in the Bombay textile trade had been reversed.[25]

British and other European firms took advantage of their connections with the Bombay government, which even the most well-established Indian firms could never enjoy.[26] This government circle influence flowed from the strength of firms' financial connections, as the familiarity of British traders with banking methods in London meant that they were less of a credit risk.[27] The British and Europeans had another advantage. They adapted to new technologies and new ways. When new methods of conducting long-distance trade replaced the old ways, British traders made the necessary adjustments. Reliable telegraphic communications increased the capabilities of the British to engage in the cotton trade. Since speed, promptness, and precision mattered for completing trade transactions, the British traders became more efficient. The Indian traders' consignment system became outmoded and cumbersome.[28]

The British traders also readily changed their strategic models. Prior to the establishment of the railways, cotton had reached Bombay in loose bales. This packaging was unsuitable for transit in a railway wagon. The Great Indian Peninsular Railway, now the Central Railway, the major line connecting Bombay to the cotton growing areas such as Nagpur, introduced a graduated freight rate penalizing

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loose cotton bales and encouraging the movement of pressed cotton.[29] The merchants who wanted to use the railways had no choice but to press cotton near where it was grown, so that it was ready for immediate export on arrival in Bombay. The Indian merchants, in Bombay and elsewhere, were slow in adopting this practice. They forfeited the inland trade to British traders. By moving inland, the British vertically integrated their operations and eliminated various middlemen.[30] This gave British firms a scale advantage.[31]

A key element in late nineteenth-century development of Indian industry was the establishment of the railway and telegraph systems. Telegraph and railway lines were laid down in India to promote resource flows and transformations, as the case of backward integration in the cotton trade reveals. Nevertheless, India's heavy engineering industry was primitive. Railway engines and telegraph equipment were manufactured abroad.[32] The Indian railway network was the fifth largest in the world by 1900. The laying of railway tracks started a chain of foreseen and unforeseen results. It provided one of the foundations of industrial enterprise.[33] The railways incurred the largest capital expenditures in nineteenth-century India. They had been rapidly built up by government bodies in the 1870s, and by private subsidized contractors in the 1860s and 1880s.[34]

But the financing was substantially tilted towards British interests.[35] The financial burden was shifted to the Indians at large through a guaranteed return system on private capital by which the companies paid a minimum sum to investors in Britain even if profits were negative. This involved the payment of £50 million by 1900.[36] The return guarantee scheme had given the government the right to purchase a line after 25 years. By the 1920s, the government owned over 67 percent of the total mileage of India's railways, plus it had a small interest in all of the railways in India.[37] Yet, extremely few superior positions in the railways were staffed by Indians. Thus, the diffusion of new skills among Indians was limited while a substantial part of the income generated through railway investments was sent overseas.[38]

Early twentieth-century tipping point

The twentieth century was a time of great change in India. These numerous changes were political, social, economic, and cultural. I will cover a few topics related to the industrial development of India,

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